

Semantic Drift Across Qwen Generations: Reproducibility, Thinking Mode, Serving Stack, and What Cosine Similarity Misses in Iterated Paraphrase Chains

James H. Smith
Joshua.AI LLC
jim@joshua8.ai

September 2026

Abstract

We extend the telephone-game study of semantic drift in iterated LLM paraphrase chains [Smith, 2026] to the Qwen3.6 and Qwen3.8 model generations, and we use the fast, cheap inference these models afford to ask questions the earlier work could not: whether a sampling seed reproduces a chain at all, what reasoning (“thinking”) mode does to drift, whether speculative decoding changes it, how prompts and drift behave across sixteen knowledge domains, and whether embedding similarity measures the loss of information that practitioners care about. Across 2,717 completed chains and 87,110 model calls on eight configurations served by vLLM, SGLang and llama.cpp, we find: (1) seeds do not pin chains on any of these stacks—identical requests diverge at the first token even at temperature 0 and even with one request in flight—so drift must be reported over samples, and we do so at 15 to 30 chains per condition; (2) thinking mode is a large effect whose sign depends on the model, costing Qwen3.6-35B up to 0.16 in final similarity and never reaching a fixed point over 100 iterations, while lifting Qwen3.8-27B from 0.58 to 0.88 on the short stimulus; no system prompt rescues the model it hurts, and it retains the runaway-thinking failure of the previous generation at a low rate—a failure that a presence penalty of 1.5 in the sampler eliminates entirely without changing the output otherwise; (3) multi-token-prediction speculative decoding leaves drift unchanged, and temperature 0 narrows the spread without changing the mean; (4) system-prompt rankings replicate across hosts ($\rho = 0.95\text{--}0.98$) and transfer across models within a generation ($\rho = 0.70\text{--}0.90$), reversing the non-transfer reported for older models, with one domain-neutral fidelity prompt raising every one of sixteen domains to 0.88–0.98; (5) most chains lock byte-for-byte within ten iterations, so “drift” on these models is largely where a copying attractor caught the chain; and (6) across sixteen domains with ten checkable facts each, cosine similarity is only weakly related to the survival of those facts ($\rho = 0.32$), and on construction statements of work it is negatively related ($\rho = -0.42$): outputs inflate two- to seven-fold and reframe as memos while every number survives. Dates are the most fragile fact type (35% survival), identifiers the most robust (81%). We conclude that similarity-based drift scores measure genre and length change far more than information loss, and we release the full chain texts so that other metrics can be applied.

1 Introduction

In the companion paper [Smith, 2026] (hereafter Paper A) we measured semantic drift in iterated paraphrase chains across seven model families and, within the Qwen3.5 family, across size, precision, quantization format, serving stack, temperature and system prompt. That study reached three conclusions that this one takes as its starting point: model choice dominates every other factor;

a discrete *runaway thinking* failure, in which a model with reasoning enabled spends its whole generation budget inside the thinking block, removes a large fraction of chains on some stacks; and system-prompt rankings did not transfer between the models tested.

Paper A also left questions it could not answer. Its experiments were run at five “seeds” per condition on the assumption that a seed reproduces a chain, which was never tested. Thinking mode could not be switched per request on the serving stacks then in use, so its effect was inferred from failure rates rather than measured. Speculative decoding was not available. Every stimulus was a cooking recipe. And cosine similarity to the original text was the only measure of drift, validated against a second embedding model but never against the information the text actually carried.

The Qwen3.6 and Qwen3.8 generations, released between April and August 2026, make those questions tractable. Served in NVFP4 on Blackwell GPUs they paraphrase a recipe in one to three seconds, thinking is a per-request switch on every serving stack, and llama.cpp exposes the same weights with and without a multi-token-prediction draft head. We therefore ran roughly three times as many chains as Paper A, at 15 to 30 samples per condition, on eight configurations of four models, and added two kinds of stimulus the earlier work lacked: construction statements of work with internally consistent quantities, and sixteen prompts from sixteen knowledge domains, each carrying ten checkable anchor facts.

Our contributions are:

1. A direct test of seed reproducibility on three serving stacks, finding none (Section 4.2), and a study design that treats every chain as an independent sample.
2. A cross-generation comparison at matched size from Qwen3 through Qwen3.5, 3.6 and 3.8, showing that drift on the short stimulus has *worsened* with each generation as models grew more verbose, while drift on the long stimulus improved (Section 4.1).
3. The first controlled measurement of thinking mode in iterated generation: a model-dependent sign, no fixed point at 100 iterations, no rescue by prompting, and a residual runaway rate (Section 4.3).
4. Null results for speculative decoding and for greedy decoding as drift controls, and a replication of the format-and-stack effect on two new generations (Sections 4.4 and 4.5).
5. Evidence that system-prompt rankings are a stable, transferable property within a generation, and that a single fidelity prompt works in every domain tested (Sections 4.6 and 4.9).
6. Byte-level locking as the dominant terminal state of a chain (Section 4.7).
7. Two new fidelity metrics—numeric-token survival and anchor-fact survival—and the finding that embedding similarity tracks neither (Sections 4.8 and 4.9).

2 Related Work

Paper A reviews the literature on iterated rephrasing and cultural attractors [Perez et al., 2025, Brinkmann et al., 2023, Acerbi and Stubbersfield, 2023], the broken-telephone effect [Mohamed et al., 2025], attractor dynamics [Wang et al., 2025, Chytas and Singh, 2026], agent drift [Rath, 2026], model collapse under recursive generation [Shumailov et al., 2024, 2023], iterated learning [Kirby et al., 2008], and quantization effects [Han et al., 2026, Hooper et al., 2024, Liu et al., 2024]. Mollah et al. [2025] apply a telephone-game protocol to unified vision-language models, alternating captioning and image generation, and find catastrophic rather than gradual collapse; their object-level compliance

score is the visual analogue of the anchor-fact survival metric introduced here. Belem et al. [2026] show that repeated rewriting hardens hedged statements into assertions while leaving surface content intact, a change no embedding metric detects; our fact-survival results are a second demonstration that similarity and fidelity come apart.

Three bodies of work bear on the new questions. Nondeterminism in LLM inference is now understood to arise from batch-dependent kernel scheduling rather than from sampling [He and Thinking Machines Lab, 2025], with batch-invariant kernels [Zhang et al., 2025b, Zhao et al., 2026] proposed as remedies; Section 4.2 measures the consequence for iterated generation. Speculative decoding [Leviathan et al., 2023, Chen et al., 2023] and its multi-token-prediction variant [Gloeckle et al., 2024, DeepSeek-AI, 2024] are distribution-preserving in principle; Section 4.4 tests whether that holds for drift in practice. Instruction-following capacity [Jaroslawicz et al., 2025] bears on why thinking mode and constraining prompts interact as they do.

3 Methodology

3.1 Framework and metrics

The chain protocol is unchanged from Paper A: a stimulus x_0 is sent with the instruction “Paraphrase this to the next agent: x_t ”, the reply becomes x_{t+1} , and cosine similarity between the Qwen3-Embedding-0.6B embeddings [Zhang et al., 2025a] of x_t and x_0 is recorded at every step. Chains run 30 iterations unless stated (100 in the long-horizon runs). Temperature is 0.7 and the generation budget 16,384 tokens unless varied. An empty reply keeps the previous message as the next input and is recorded.

Four per-chain measures are new. *Locking*: an iteration whose output is byte-identical to its input; a chain is *locked at end* if its last three outputs are identical. *Inflation*: final word count divided by the stimulus word count. *Meta-commentary*: whether the final output contains framing phrases such as “here is the paraphrase” or “next agent”. *Fact survival*: for the construction documents, the fraction of the stimulus’s numeric tokens present verbatim in the output; for the breadth prompts, the fraction of ten hand-chosen anchor facts (numbers with units, names, dates, identifiers) present after whitespace and case normalisation. All are computed offline from the saved texts.

3.2 Models and serving

Table 1 lists the eight configurations. The NVFP4 builds [NVIDIA et al., 2025, NVIDIA, 2026] run on vLLM [Kwon et al., 2023] for the 35B mixture-of-experts and 27B dense models and on SGLang [Zheng et al., 2024] for Flash-Next, all with an fp8 KV cache and thinking disabled by default; thinking is enabled per request through the chat-template switch and the reasoning block is separated from the returned content by the server’s reasoning parser. The GGUF builds [Gerganov and llama.cpp contributors, 2023, Unsloth, 2026a,b] run on llama.cpp with a q8_0 KV cache and reasoning disabled; the Qwen3.8-27B GGUF file carries an embedded multi-token prediction head that llama.cpp can use as a self-speculative draft, giving a same-weights comparison with and without speculative decoding. Two byte-identical vLLM deployments of Qwen3.6-35B on different machines (hosts A and B) provide a cross-host replicate.

Table 1: Model inventory. All Qwen builds are the instruct checkpoints [Qwen Team, 2026b,a,c,d]. GGUF tier is Unsloth UD-Q4_K_XL. Hardware is given in Appendix C.

Model	Architecture	Format	Stack	Thinking switchable
Qwen3.6-35B-A3B (hosts A, B)	MoE, 3B active	NVFP4	vLLM	yes
Qwen3.6-35B-A3B	MoE, 3B active	GGUF Q4_K_XL	llama.cpp	no (off)
Qwen3.6-27B	dense	GGUF Q4_K_XL	llama.cpp	no (off)
Qwen3.8-27B	dense, hybrid attention	NVFP4	vLLM (TP2)	yes
Qwen3.8-27B	dense	GGUF Q4_K_XL	llama.cpp	no (off)
Qwen3.8-27B + MTP	dense	GGUF Q4_K_XL	llama.cpp, draft-MTP	no (off)
Qwen3.8-Flash-Next	MoE 125B, 6B active	NVFP4	SGLang	yes

3.3 Stimuli

Three stimulus sets are used. The two recipes of Paper A (3-step *spaghetti*, 21-step *lasagna*, reproduced in Appendix A) anchor every comparison with the earlier data. Two construction statements of work in the format that a specification ingestion product emits for subcontractor bidding—a 95-word summary with three scope items and one exclusion, and a 585-word detailed statement with twenty scope items, an exclusions list attributing work to other trades, an alternate and a base bid—carry 20 and 131 numeric tokens respectively, with every volume computed from the stated dimensions so that a drifted number is detectable. Sixteen prompts of 145–190 words, one per domain (recipe, laboratory protocol, lockout/tagout procedure, concrete scope of work, construction RFI, change order, contract clause, building code, hospital discharge instructions, Python function, REST API reference, scientific abstract, financial summary, news report, travel itinerary and literary prose), each carry exactly ten anchor facts tagged by type. Appendix A gives every text and anchor.

3.4 Experiments

All conditions use 15 samples unless stated; a sample is one chain. Table 2 and the per-section tables carry n .

1. **Baseline:** both recipes on all eight configurations; 30 samples on the pooled Qwen3.6-35B NVFP4 hosts.
2. **Reproducibility:** the same ten seeds run twice on host A, once on host B, and twice more on host A with one request in flight; 15 samples at temperature 0 on the three GPU models.
3. **Thinking:** both recipes with thinking on, on the four thinking-capable deployments; 100-iteration runs with thinking off (host B, 15 samples) and on (host B 15, Qwen3.8-27B and Flash-Next 5 each); thinking combined with all 20 system prompts on host A (*lasagna*, 5 samples).
4. **Format and stack:** NVFP4 against GGUF for Qwen3.6-35B and Qwen3.8-27B; Qwen3.8-27B GGUF with and without MTP.
5. **Temperature:** $T \in \{0, 0.3, 0.5, 0.7, 1.0\}$ on the three GPU models and both hosts, 5 samples per point except 15 at $T = 0$ and $T = 0.7$.
6. **System prompts:** the 20 prompts of Paper A on both recipes for Qwen3.6-35B (both hosts), Qwen3.8-27B and Flash-Next, 5 samples (10 for host A *spaghetti*).

7. **Construction:** both statements of work, thinking off and on, on the three GPU models (30 samples for the pooled 35B hosts, 15 otherwise), plus the six domain-neutral system prompts on host B.
8. **Breadth:** the sixteen domain prompts with thinking off on Qwen3.6-35B (15 samples per host, pooled to 30) and Flash-Next (5), with thinking on (host B, 5) and with the Ultra-Constrained Fidelity prompt (host A, 5).
9. **Sampler:** Qwen3.6-35B on host A with thinking on, lasagna with the Visual Tutorial Style prompt (the condition with the highest token-cap rate), 15 samples under each of three samplers: the server default (top- k 20, top- p 0.95, no penalty, read from the model card), a bare sampler (top- k off, top- p 1.0, no penalty) and the vendor’s thinking-mode recommendation (top- k 20, top- p 0.95, presence penalty 1.5); plus the bare sampler on plain spaghetti as a control.

In total 2,717 chains and 87,110 model calls completed with no API errors. Every thinking-off chain produced text at every iteration; the empty completions in the thinking-on conditions are analysed in Section 4.3.

3.5 Statistics

Every mean carries a percentile bootstrap 95% interval ($B = 10,000$) [Efron and Tibshirani, 1993]. Two-group contrasts use the Mann–Whitney U test, rank comparisons Spearman’s ρ with a bootstrap interval, and rates a Wilson interval. Because Section 4.2 shows that seeds do not reproduce chains, no test treats seed-matched chains as pairs.

4 Results

4.1 Baseline drift and the cross-generation picture

Figure 1 and Table 2 give the baseline. On the 21-step lasagna recipe every configuration finishes between 0.85 and 0.91, a narrow band with Qwen3.8-27B NVFP4 (0.905) and the MTP build (0.897) at the top and Qwen3.6-35B GGUF (0.847) at the bottom. On the 3-step spaghetti recipe the configurations spread from 0.53 to 0.74 and their within-condition spread is three to eight times larger (standard deviations of 0.09–0.25 against 0.02–0.07), with individual chains as low as 0.12. The short stimulus, not the long one, is the hard case for these models, and the reason is visible in the last two columns of the table: on spaghetti every configuration inflates the 17-word recipe five- to seven-fold and attaches meta-commentary in 80–100% of chains, whereas on lasagna the word count is preserved to within 10%.

Across generations. Figure 2 and Table 3 place these results beside Paper A’s Qwen3 pilot and Qwen3.5 data at matched size, using Paper A’s clean chains (those without runaway failures) for the Qwen3.5 rows. On the short stimulus the trend is monotonically downward after Qwen3.5: the 27B dense line runs 0.564 (Qwen3-32B, single pilot chain), 0.914 (Qwen3.5), 0.680 (Qwen3.6), 0.594 (Qwen3.8), and the 35B-class MoE line 0.806, 0.923, 0.576 (GGUF) or 0.736 (NVFP4); the 122B class falls from 0.925 (Qwen3.5-122B) to 0.731 (Flash-Next). On the long stimulus the newer generations are as good or better: 0.829 (Qwen3.5-27B) against 0.889 and 0.878–0.905, and 0.812–0.845 against 0.847–0.896 for the 35B class. The Qwen3.5 short-stimulus values rest on two to three clean chains each and are existence results, as Paper A notes; the newer values rest on fifteen to thirty. The pattern is nonetheless consistent with everything else in this paper: the newer models paraphrase a long document faithfully and cannot leave a short one alone.

F1. Baseline drift trajectories (mean $\pm$ bootstrap 95% CI)

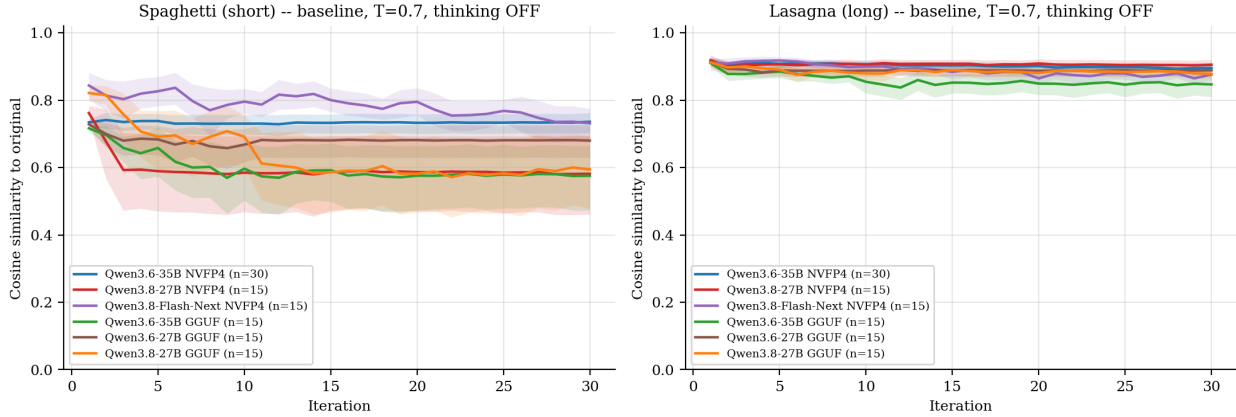


Figure 1: Baseline drift trajectories for the six non-MTP configurations, mean with bootstrap band; n in the legend. Left: spaghetti. Right: lasagna.

Table 2: Baseline drift at iteration 30 ($T=0.7$, no system prompt, thinking OFF). Means over independent chains with percentile bootstrap 95% CIs ($B=10,000$). Locked = last three outputs byte-identical; inflation = final output words divided by prompt words; meta = final output contains agent-handoff / meta-commentary markers. The two Qwen3.6-35B NVFP4 hosts are pooled.

Configuration	Recipe	Mean	95% CI	n	Locked	Inflation	Meta
Qwen3.6-35B NVFP4	Spaghetti	0.736	[0.703, 0.763]	30	0.87	5.64	1.00
Qwen3.6-35B NVFP4	Lasagna	0.896	[0.878, 0.910]	30	0.87	1.02	0.83
Qwen3.8-27B NVFP4	Spaghetti	0.582	[0.458, 0.690]	15	0.93	7.21	1.00
Qwen3.8-27B NVFP4	Lasagna	0.905	[0.891, 0.919]	15	0.93	1.05	0.87
Qwen3.8-Flash-Next NVFP4	Spaghetti	0.731	[0.675, 0.777]	15	0.20	5.71	0.80
Qwen3.8-Flash-Next NVFP4	Lasagna	0.876	[0.859, 0.892]	15	0.00	1.03	0.60
Qwen3.6-35B GGUF	Spaghetti	0.576	[0.472, 0.668]	15	0.73	6.83	1.00
Qwen3.6-35B GGUF	Lasagna	0.847	[0.810, 0.876]	15	0.93	0.96	0.80
Qwen3.6-27B GGUF	Spaghetti	0.680	[0.629, 0.726]	15	0.93	7.19	1.00
Qwen3.6-27B GGUF	Lasagna	0.889	[0.873, 0.905]	15	1.00	1.01	0.87
Qwen3.8-27B GGUF	Spaghetti	0.594	[0.481, 0.695]	15	0.60	6.17	0.93
Qwen3.8-27B GGUF	Lasagna	0.878	[0.863, 0.891]	15	0.87	1.10	1.00
Qwen3.8-27B GGUF+MTP	Spaghetti	0.532	[0.411, 0.648]	15	0.73	6.85	0.93
Qwen3.8-27B GGUF+MTP	Lasagna	0.897	[0.885, 0.909]	15	0.80	1.07	0.87

4.2 Seeds do not reproduce chains

Every experiment in Paper A, and in the prior literature we are aware of, labels repeated runs by sampling seed on the assumption that the seed fixes the trajectory. Figure 3 and Table 4 test that assumption on host A. Ten seeds run twice produced ten pairs of chains that differed at iteration 1 in every case, with zero identical iterations in any pair and final similarities differing by 0.045 on average (maximum 0.096). Running the same five seeds twice with a single request in flight, so that no other request could share a batch, changed nothing: all five pairs diverged at iteration 1. Two byte-identical single requests at temperature 0 with the same seed returned different text, disagreeing

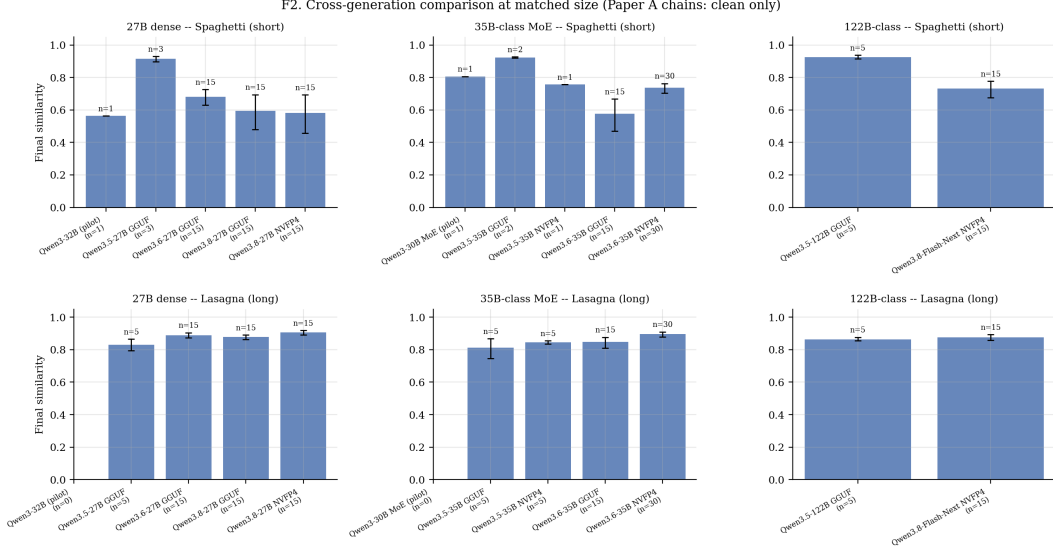


Figure 2: Final similarity at matched model size across four Qwen generations. Qwen3 values are single unseeded pilot chains and have no lasagna run; Qwen3.5 values are Paper A’s clean chains only, with n shown. Error bars are bootstrap 95% intervals.

at about the ninth token. The deployment—an NVFP4 mixture-of-experts on a FlashInfer backend with prefix caching—is not batch-invariant in the sense of He and Thinking Machines Lab [2025], and neither the seed nor greedy decoding restores reproducibility.

What does replicate is the distribution. Host A and host B, two identical deployments on different machines, give baseline means that are statistically indistinguishable (spaghetti 0.749 vs. 0.723, $p = 0.63$; lasagna 0.904 vs. 0.887, $p = 0.38$), and every ranking reported later in this paper agrees across the two hosts at $\rho \geq 0.95$. Temperature 0 does narrow the spread: at 15 samples per condition the standard deviation of final similarity falls from 0.089 to 0.016 on Qwen3.6-35B spaghetti and from 0.240 to 0.070 on Qwen3.8-27B spaghetti, but the means move by less than 0.05 in five of six cases (the exception is Flash-Next on lasagna, 0.918 against 0.876, $p < 0.001$). Greedy decoding is a variance reduction, not a fidelity improvement, and it is not reproducible either.

The consequence for method is that a “seed” is one draw from the model’s output distribution under that deployment. We therefore report every condition as a sample of 15 to 30 chains with a bootstrap interval, we never pair chains by seed, and we regard the roughly 0.05 within-host difference between draws as the noise floor below which a contrast between conditions is not interpretable.

4.3 Thinking mode

Table 5 and Figure 4 give the central result of this section: enabling reasoning changes drift by 0.03 to 0.30 in every condition, significant in all eight, and the direction is a property of the model. Qwen3.6-35B loses 0.09 on spaghetti and 0.15–0.16 on lasagna, identically on both hosts, while generating 1,400–3,100 reasoning tokens per iteration. Qwen3.8-27B gains 0.30 on spaghetti and 0.03 on lasagna with only 110–440 reasoning tokens, and Flash-Next gains 0.14 and 0.06 with 95–310. Thinking also collapses the variance of the model it helps: Qwen3.8-27B’s spaghetti standard deviation falls from 0.24 to 0.02, because the reasoning step suppresses the failure that dominates its baseline, in which the model answers the paraphrase request with a menu of three options, the

Table 3: Cross-generation drift at matched model size. Paper A rows use clean chains only (no empty output in any successful iteration); the Qwen3 pilot is a single unseeded chain per model. Bootstrap 95% CIs, $B=10,000$.

Model	Spaghetti			Lasagna	
	Mean	95% CI	n	Mean	95% CI
<i>27B dense</i>					
Qwen3-32B (pilot)	0.564	[0.564, 0.564]	1	—	—
Qwen3.5-27B GGUF	0.914	[0.898, 0.929]	3	0.829	[0.793, 0.866]
Qwen3.6-27B GGUF	0.680	[0.630, 0.727]	15	0.889	[0.872, 0.905]
Qwen3.8-27B GGUF	0.594	[0.480, 0.694]	15	0.878	[0.864, 0.891]
Qwen3.8-27B NVFP4	0.582	[0.456, 0.693]	15	0.905	[0.891, 0.919]
<i>35B-class MoE</i>					
Qwen3-30B MoE (pilot)	0.806	[0.806, 0.806]	1	—	—
Qwen3.5-35B GGUF	0.923	[0.920, 0.927]	2	0.812	[0.746, 0.867]
Qwen3.5-35B NVFP4	0.757	[0.757, 0.757]	1	0.845	[0.836, 0.854]
Qwen3.6-35B GGUF	0.576	[0.470, 0.667]	15	0.847	[0.811, 0.876]
Qwen3.6-35B NVFP4	0.736	[0.703, 0.763]	30	0.896	[0.878, 0.910]
<i>122B-class</i>					
Qwen3.5-122B GGUF	0.925	[0.915, 0.937]	5	0.864	[0.856, 0.877]
Qwen3.8-Flash-Next NVFP4	0.731	[0.675, 0.777]	15	0.876	[0.859, 0.892]

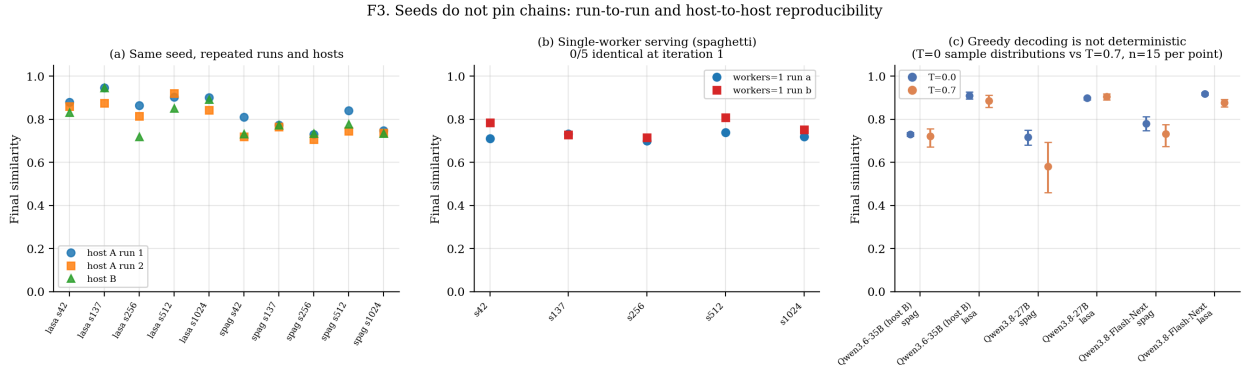


Figure 3: Seed reproducibility. (a) Final similarity for ten seeds run twice on host A and once on host B. (b) The same for five seeds with a single request in flight. (c) Distributions of final similarity at $T = 0$ and $T = 0.7$, 15 samples each, three models.

next iteration paraphrases the menu, and the chain locks into a memo about a missing source text at similarity 0.14.

No fixed point. With thinking off, 100-iteration chains on host B are flat after iteration 30: the mean moves by -0.008 (spaghetti) and -0.005 (lasagna) between iterations 30 and 100, and

Table 4: Seed reproducibility. Chains repeated with identical seed, prompt and deployment diverge immediately; “identical at iter. 1” counts pairs whose first outputs are byte-identical, and $|\Delta|$ is the absolute difference in final similarity. The lower block compares the two Qwen3.6-35B NVFP4 hosts as distributions (Mann–Whitney U, $n=15$ per host per recipe).

Comparison	pairs	identical at iter. 1	mean identical iters	mean $ \Delta $	max $ \Delta $
host A run1 vs run2	10	0/10	0.00	0.045	0.096
host A run1 vs host B	10	1/10	3.00	0.041	0.144
host A workers=1 run a vs run b	5	0/5	0.00	0.040	0.074

Recipe	host A mean (n)	host B mean (n)	MWU p
spaghetti	0.749 (15)	0.723 (15)	0.6333
lasagna	0.904 (15)	0.887 (15)	0.3837

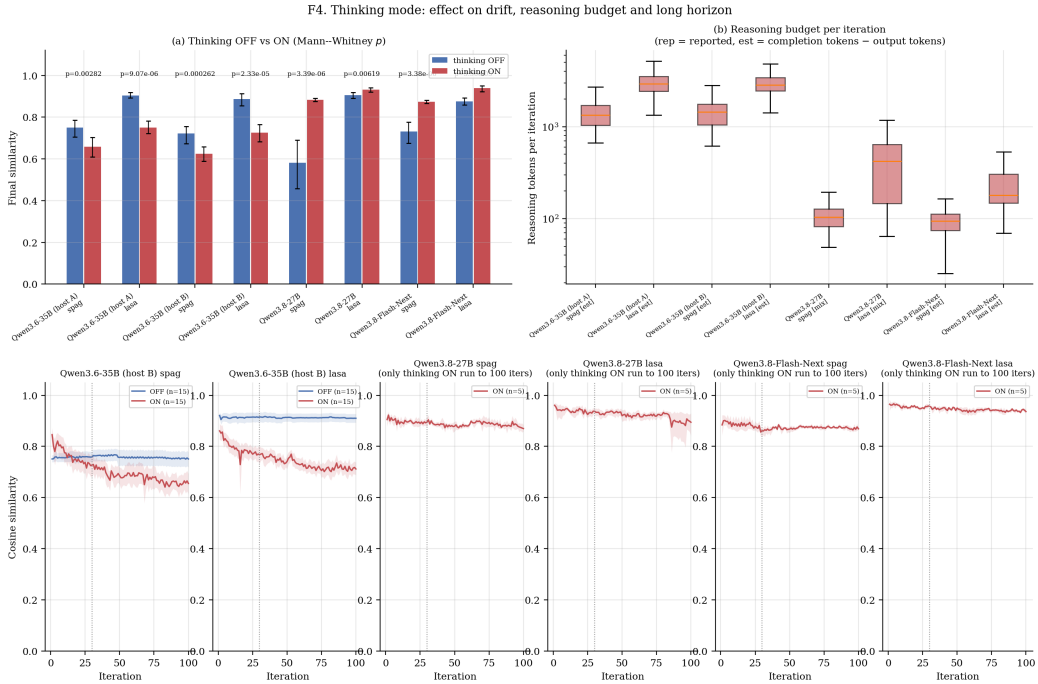


Figure 4: Thinking mode. (a) Final similarity with thinking off and on, per configuration and recipe. (b) Reasoning tokens per iteration. (c) 100-iteration trajectories with thinking off and on.

only one to two of fifteen chains dip by more than 0.05 in that span. With thinking on the same model keeps drifting: -0.072 and -0.056 over the same span, with 73% and 100% of chains dipping, reaching 0.56 and 0.63 at their minima. Qwen3.8-27B with thinking drifts mildly over 100 iterations and Flash-Next holds. A reasoning model that rewrites a text is, on this evidence, not an iterated function with a fixed point; each pass reconsiders the document.

Table 5: Thinking mode. Upper block: final similarity with thinking OFF and ON (mean with n chains), the ON–OFF difference, Mann–Whitney p , and the mean reasoning tokens per iteration. “rep” = reasoning token counts reported by the server; “est” = completion tokens minus output tokens (4 characters per token). Lower block: 100-iteration runs, similarity at iteration 30 and 100 and the fraction of chains dipping more than 0.05 below their iteration-30 value.

Configuration	Recipe	OFF (n)	ON (n)	Δ	p	Reas. tok/iter	Source
Qwen3.6-35B NVFP4 (host A)	spaghetti	0.749 (15)	0.658 (15)	-0.091	0.0028	1420	estimated
Qwen3.6-35B NVFP4 (host A)	lasagna	0.904 (15)	0.750 (15)	-0.154	<0.0001	3095	estimated
Qwen3.6-35B NVFP4 (host B)	spaghetti	0.723 (15)	0.625 (15)	-0.097	0.0003	1466	estimated
Qwen3.6-35B NVFP4 (host B)	lasagna	0.887 (15)	0.726 (15)	-0.160	<0.0001	2975	estimated
Qwen3.8-27B NVFP4	spaghetti	0.582 (15)	0.883 (15)	0.302	<0.0001	110	estimated/reported
Qwen3.8-27B NVFP4	lasagna	0.905 (15)	0.932 (15)	0.026	0.0062	436	estimated/reported
Qwen3.8-Flash-Next NVFP4	spaghetti	0.731 (15)	0.873 (15)	0.141	<0.0001	95	estimated
Qwen3.8-Flash-Next NVFP4	lasagna	0.876 (15)	0.939 (15)	0.063	<0.0001	306	estimated

Configuration	Recipe	Think	n	@30	@100	$\Delta_{30 \rightarrow 100}$	dip rate
Qwen3.6-35B NVFP4 (host B)	spaghetti	OFF	15	0.759	0.751	-0.008	0.07
Qwen3.6-35B NVFP4 (host B)	spaghetti	ON	15	0.727	0.655	-0.072	0.73
Qwen3.6-35B NVFP4 (host B)	lasagna	OFF	15	0.915	0.910	-0.005	0.13
Qwen3.6-35B NVFP4 (host B)	lasagna	ON	15	0.767	0.712	-0.056	1.00
Qwen3.8-27B NVFP4	spaghetti	ON	5	0.895	0.870	-0.025	0.40
Qwen3.8-27B NVFP4	lasagna	ON	5	0.938	0.894	-0.043	0.60
Qwen3.8-Flash-Next NVFP4	spaghetti	ON	5	0.861	0.869	0.008	0.00
Qwen3.8-Flash-Next NVFP4	lasagna	ON	5	0.957	0.936	-0.021	0.20

No prompt rescues it. On Qwen3.6-35B lasagna with all 20 system prompts, thinking on is worse than thinking off for 18 of the 20 (mean -0.053), the best prompt with thinking (Precise Transmitter, 0.915) merely matches the no-prompt baseline without thinking (0.913), and the prompt ranking with thinking correlates only weakly with the ranking without ($\rho = 0.43$). Where thinking helps, in Qwen3.8-27B, it does so because a short reasoning pass suppresses a formatting failure; where it hurts, the reasoning pass rewrites the document from its own summary of it, and the prompt cannot reach that summary.

Runaway thinking persists at a low rate. Paper A found that Qwen3.5 with reasoning enabled exhausted its 16,384- token budget inside the thinking block in 13.5% of chains. Here every one of the 2,222 thinking-off chains produced text at every iteration (runaway rate 0, Wilson interval $[0, 0.17\%]$), and so did every thinking-on chain on Qwen3.8-27B and Flash-Next. On Qwen3.6-35B with thinking on, 17 of the 300 such chains (5.7%) contained at least one iteration that hit the budget, concentrated in the lasagna system-prompt conditions and the long construction document, and 13 of those returned an empty completion. The failure Paper A described has therefore been reduced by an order of magnitude in the following generation and removed in the one after, but it has not disappeared, and the prompts that trigger it (Visual Tutorial Style, Friendly Conversational, Multimodal Visual Focus) are the verbose ones rather than, as in Qwen3.5, the constraining ones.

The sampler controls it. Paper A could not separate the serving stack from the sampling defaults each stack applies, because its requests set only temperature and seed. We therefore took

the worst condition above—Visual Tutorial Style on lasagna, where the server default sampler hit the token cap in 8 of 15 chains—and reran it under two explicit samplers. With top- k and top- p removed (a bare sampler) the cap was hit in 5 of 15 chains, not distinguishable from the default ($p = 0.46$, Fisher). With the vendor’s recommended thinking-mode sampler, which adds a presence penalty of 1.5, the cap was hit in 0 of 15 chains ($p = 0.002$ against the default, $p = 0.04$ against the bare sampler), the longest single reply fell from the 16,384 cap to 14,225 tokens, and mean reply length fell by 8%. Final similarity was unchanged (0.56, 0.58, 0.59; $p = 0.74$). The bare sampler on plain spaghetti, where the default rate is zero, stayed at zero. Runaway thinking is therefore not a property of a serving stack but of sampling the reasoning block without a repetition control, which is exactly the setting the Qwen model cards warn against; a presence penalty removes the failure without changing what the model produces when it does answer. This also resolves the stack ordering Paper A reported: the Ollama library’s Qwen3.5 tags ship a parameter layer with exactly this sampler (top- k 20, top- p 0.95, presence penalty 1.5), so Paper A’s Ollama arm ran penalised and its SGLang arm did not, and the Ollama arm’s lower but non-zero rate (15–27% against 36–54%) shows that on Qwen3.5 the penalty mitigates the failure where on Qwen3.6 it removes it.

4.4 Serving format, stack, and speculative decoding

Paper A found that 4-bit builds of the same Qwen3.5 weights differed by up to 0.26 on the short stimulus depending on format and serving stack. Table 6 and Figure 5 repeat the comparison on two new generations, with the caveat that the GGUF arm here differs from the NVFP4 arm in quantization, serving software, KV-cache precision and GPU generation at once (Appendix C). For Qwen3.6-35B the NVFP4 build on vLLM is better on both recipes (0.736 vs. 0.576 on spaghetti, $p < 0.001$; 0.896 vs. 0.847 on lasagna, $p = 0.006$). For Qwen3.8-27B the two builds are indistinguishable on spaghetti (0.582 vs. 0.594, $p = 0.93$) and the NVFP4 build is slightly better on lasagna (0.905 vs. 0.878, $p = 0.016$). The direction of Paper A’s effect, which favoured GGUF, is not reproduced; its magnitude on one of the two models is.

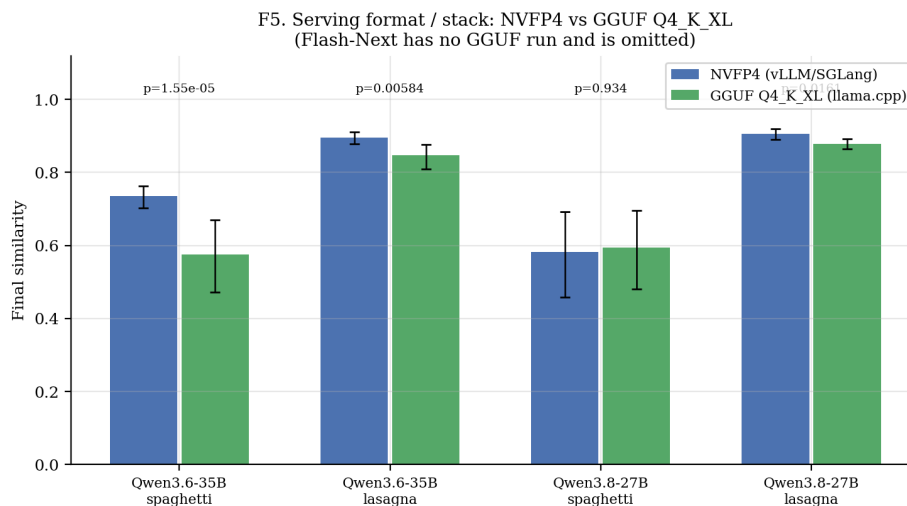


Figure 5: NVFP4 on vLLM against GGUF Q4.K.XL on llama.cpp for the same weights, both recipes, with bootstrap intervals.

Table 6: Serving format and stack. Same weights family, two quantisation formats and two inference stacks (NVFP4 on vLLM/SGLang vs GGUF Q4_K_XL on llama.cpp). Qwen3.8-Flash-Next has no GGUF run and is omitted.

Model	Recipe	NVFP4 mean [CI]	n	GGUF mean [CI]	n	Δ	MWU	p
Qwen3.6-35B	spaghetti	0.736 [0.703, 0.763]	30	0.576 [0.471, 0.670]	15	0.160	<0.0001	
Qwen3.6-35B	lasagna	0.896 [0.878, 0.911]	30	0.847 [0.810, 0.876]	15	0.049	0.0058	
Qwen3.8-27B	spaghetti	0.582 [0.457, 0.693]	15	0.594 [0.481, 0.696]	15	-0.013	0.9339	
Qwen3.8-27B	lasagna	0.905 [0.891, 0.919]	15	0.878 [0.864, 0.892]	15	0.027	0.0161	

Speculative decoding is a null. The Qwen3.8-27B GGUF file served with and without its multi-token-prediction draft head is the cleanest infrastructure comparison in either paper: same bytes, same server, one flag. Figure 6 shows no difference (spaghetti 0.594 vs. 0.532, $p = 0.44$; lasagna 0.878 vs. 0.897, $p = 0.12$; 15 samples each). Draft-and-verify sampling is distribution-preserving in theory [Leviathan et al., 2023, Chen et al., 2023], and for drift it is so in practice. An earlier five-sample reading of the same comparison had suggested a large penalty; it did not survive replication, which is the point of Section 4.2.

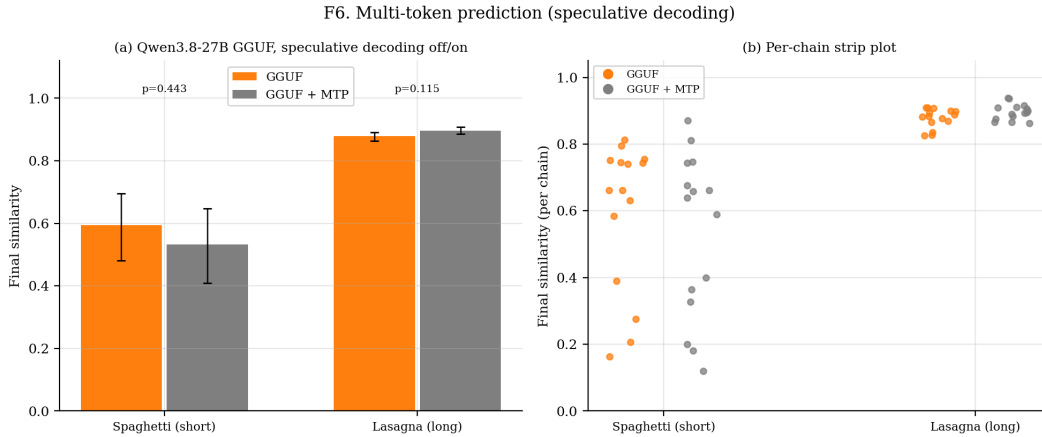


Figure 6: Multi-token-prediction speculative decoding on Qwen3.8-27B GGUF: per-chain final similarity with and without the draft head.

4.5 Temperature

Figure 7 extends the temperature ablation to the new models with 15 samples at $T = 0$. The stable configuration, Qwen3.6-35B, is flat within 0.05 across the range on both recipes and both hosts. Qwen3.8-27B on spaghetti is U-shaped, 0.718 at $T = 0$, falling to 0.582 at $T = 0.7$ and recovering to 0.747 at $T = 1.0$, the same non-monotonic pattern Paper A reported for small Qwen3.5 models. Flash-Next declines with temperature on lasagna (0.918 to 0.792). None of the three models is improved by greedy decoding beyond the variance reduction described in Section 4.2.

4.6 System prompts transfer within a generation

Paper A found that prompt rankings did not transfer across families and transferred only partially within Qwen3.5. Figure 8 and Table 7 give a different answer for Qwen3.6 and 3.8. The effect

F7. Temperature sweep (point labels give n chains)

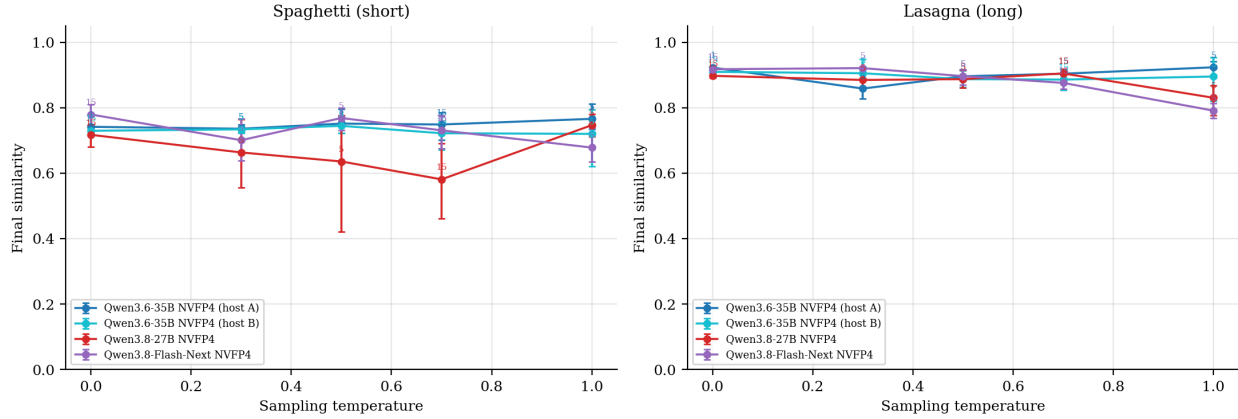


Figure 7: Final similarity against temperature; n per point in the legend. Both Qwen3.6-35B hosts are shown separately.

is large everywhere: the spread between best and worst of the 20 prompts is 0.24–0.43 on every configuration and both recipes, and the complexity threshold of Paper A, below which prompts were invisible on the short stimulus, does not exist for these models. The rankings are also stable. Across the two hosts they agree at $\rho = 0.98$ (spaghetti) and 0.95 (lasagna); across models within the same recipe at 0.83–0.90 on spaghetti and 0.70–0.88 on lasagna; and across recipes within a model at only 0.25–0.58. A prompt ranking is thus a real, measurable property of a model-and-stimulus pair, reproducible at five samples, and it is shared by the models of this generation on a given stimulus.

The winners and losers are consistent. The anti-drift and constrained categories score 0.86–0.95, the visually oriented prompts 0.65–0.78 on every configuration, below the no-prompt baseline, and the best single prompt is a fidelity instruction (100% Fidelity, Ultra-Constrained Fidelity or Precise Transmitter) on six of eight configuration–recipe pairs, at 0.94–0.97. On Qwen3.8-27B spaghetti the worst prompt is no prompt at all (0.672), because any instruction suppresses the menu-of- options failure described in Section 4.3.

4.7 Byte-level locking

Paper A described fixed-point attractors as similarity plateaus. The saved texts show something stronger. Figure 9 reports the fraction of chains whose last three outputs are byte-identical: 87% for Qwen3.6-35B NVFP4 on both recipes, 93% for Qwen3.8-27B NVFP4, 60–100% for the GGUF builds, with mean lock onset between iterations 3.5 and 9.5. At temperature 0.7 these models copy their input verbatim once it has been rewritten a few times; “paraphrase” collapses into “repeat”, and the final similarity mostly records where the copying attractor caught the chain. Flash-Next is the exception (20% on spaghetti, 0% on lasagna, onset at iteration 27 when it locks at all) and keeps rewriting to the end. With thinking on no chain of any model locks, which is the mechanism behind the missing fixed point of Section 4.3: a reasoning pass never reproduces its input exactly.

4.8 Construction statements of work: similarity against numbers

Table 8 and Figure 10 report the first non-recipe stimulus. On the 95-word summary statement of work, all three models finish at similarity 0.73–0.75, a value that in the recipe data would indicate substantial drift. Yet every one of the document’s numeric tokens survives thirty paraphrases in

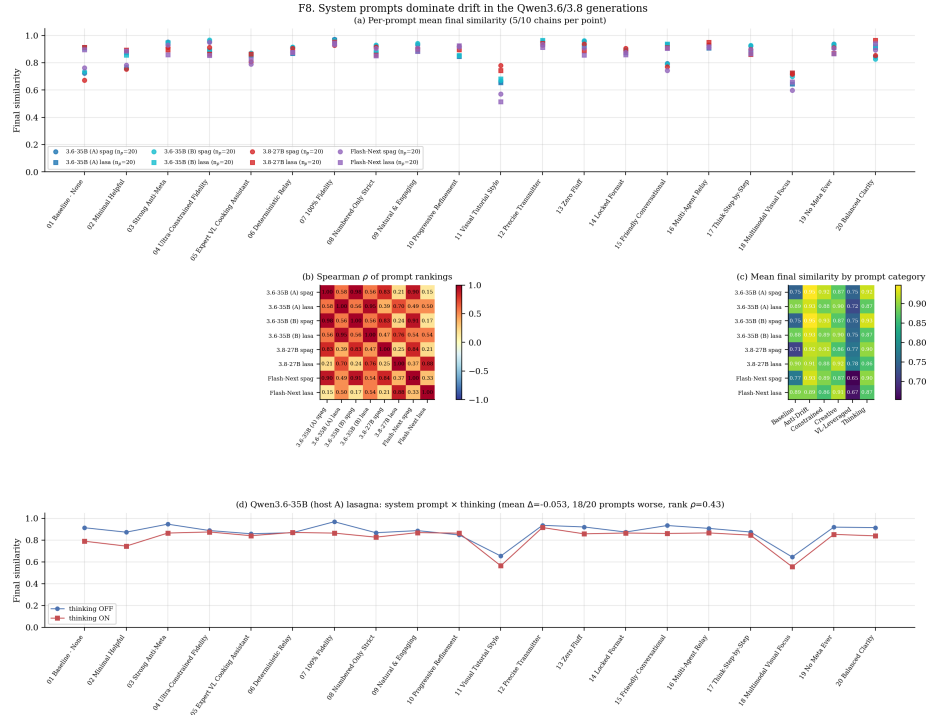


Figure 8: System prompts. (a) Per-prompt final similarity, 20 prompts, four configurations, both recipes. (b) Spearman rank correlation between configuration–recipe pairs. (c) Category means. (d) Thinking off against on per prompt, Qwen3.6-35B lasagna.

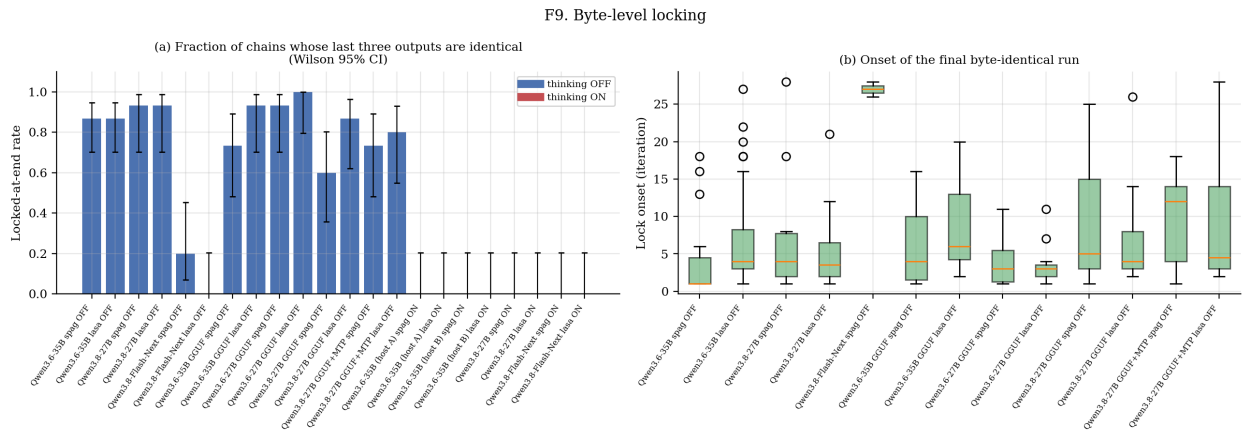


Figure 9: Locked-at-end rate with Wilson intervals, and lock-onset distribution, per configuration and recipe, thinking off and on.

essentially every chain (numeric survival 0.998–1.000). What the similarity score is registering is that the models inflate the document 2.3–2.4-fold and, in 60–100% of chains, reframe it as a memo or cover email to a project manager, with placeholders for the sender’s name; the quantities, specifications and standards are all still there. On the 585-word detailed statement, similarity is higher (0.81–0.85) while numeric survival is lower (0.85–0.91), and the exclusions list, the clause a bidder must see, is retained in 97–100% of chains. Across the 120 baseline chains, similarity and numeric survival are negatively correlated ($\rho = -0.42$, $p < 0.001$), and across all 270 concrete chains weakly so ($\rho = -0.18$). In this domain the metric points the wrong way.

Thinking mode has a domain-specific cost that the similarity score hides. With thinking on, similarity rises on every model (to 0.77–0.87), genre drift disappears on the Qwen3.8 models, and the numbers still survive (0.88–1.00); but the exclusions clause is dropped in 60–80% of short-document chains (retention 0.87 to 0.40 on Qwen3.6-35B, 0.80 to 0.20 on Qwen3.8-27B, 0.67 to 0.27 on Flash-Next). The reasoning pass tidies the document and deletes the one line whose absence changes what the reader would bid. The Ultra-Constrained Fidelity prompt, by contrast, gives 0.96 and 0.98 with every number, every exclusion and no reframing.

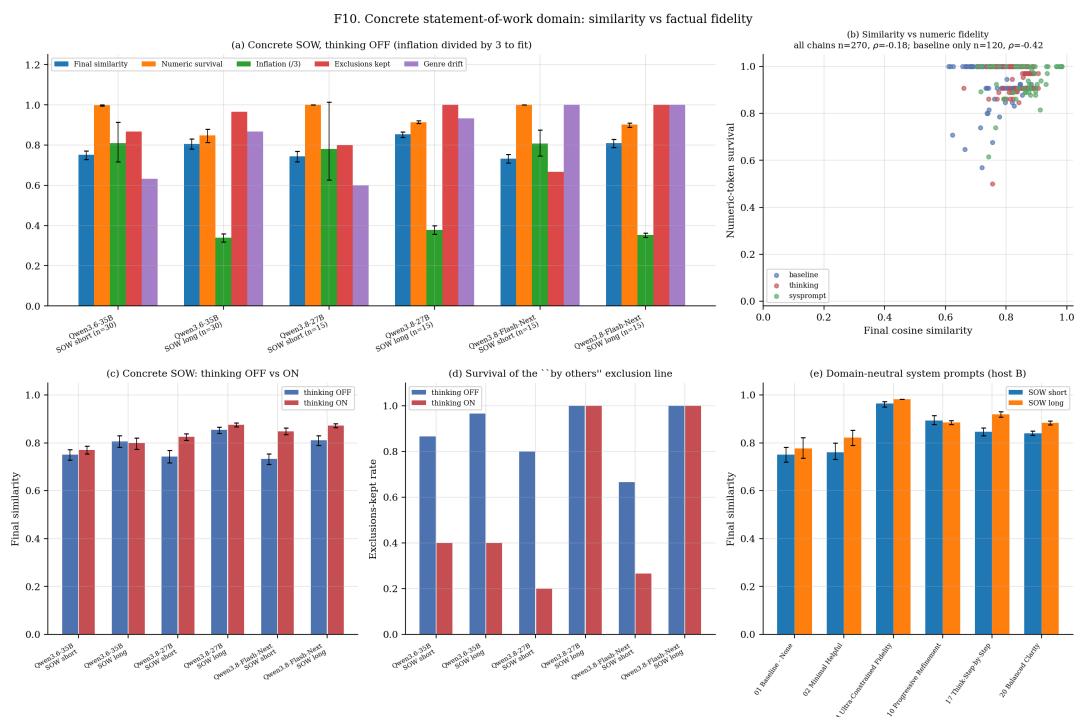


Figure 10: Construction statements of work: final similarity, numeric-token survival, inflation, exclusion retention and genre drift, thinking off and on; the six domain-neutral prompts on host B; and similarity against numeric survival per chain.

4.9 Sixteen domains: what survives and what the metric sees

Table 9, Table 10 and Figure 11 give the breadth experiment on Qwen3.6-35B at 30 samples per domain. Final similarity is remarkably insensitive to domain, spanning only 0.72 (literary prose) to 0.86 (discharge instructions), all below the lasagna recipe’s 0.90. Anchor-fact survival spans three times that range: 0.30 for the laboratory protocol and 0.33 for literary prose, 0.43 for the change order and 0.49 for the financial summary, up to 0.91–0.93 for the travel itinerary, construction RFI

and contract clause. At chain level similarity and survival correlate at $\rho = 0.32$; at domain level at 0.37 with an interval including zero. The laboratory protocol keeps 30% of its facts at similarity 0.81; the contract clause keeps 93% at 0.84. A score of 0.8 can mean that the document is intact or that most of what it said is gone.

The domain ranking is real: it replicates across hosts at $\rho = 0.95$ on both metrics and transfers to Flash-Next at 0.77–0.78. By fact type (Table 10), identifiers (0.81) and names (0.80) survive best, numbers with units worse (0.63), and dates worst of all (0.35): the financial summary lost every date in every chain and the change order kept 13%. Meta-commentary framing is present in 70–97% of final outputs in every domain, inflation runs 1.1–1.9 fold, and 53–93% of chains end locked.

Thinking on has no net effect across domains (0.772 against 0.792, $p = 0.29$) but reduces anchor survival (0.60 against 0.68) and is destructive on the two domains where it matters most, literary prose (−0.20) and the recipe (−0.15). The Ultra-Constrained Fidelity prompt, one domain-neutral sentence, raises similarity to 0.95 and anchor survival to 0.97 across the sixteen domains, improving every one of them ($p < 10^{-6}$), with inflation 1.05 and meta-commentary in 6% of outputs. A 30-step paraphrase chain on this model is close to lossless in every domain tested when told, in plain terms, to be.

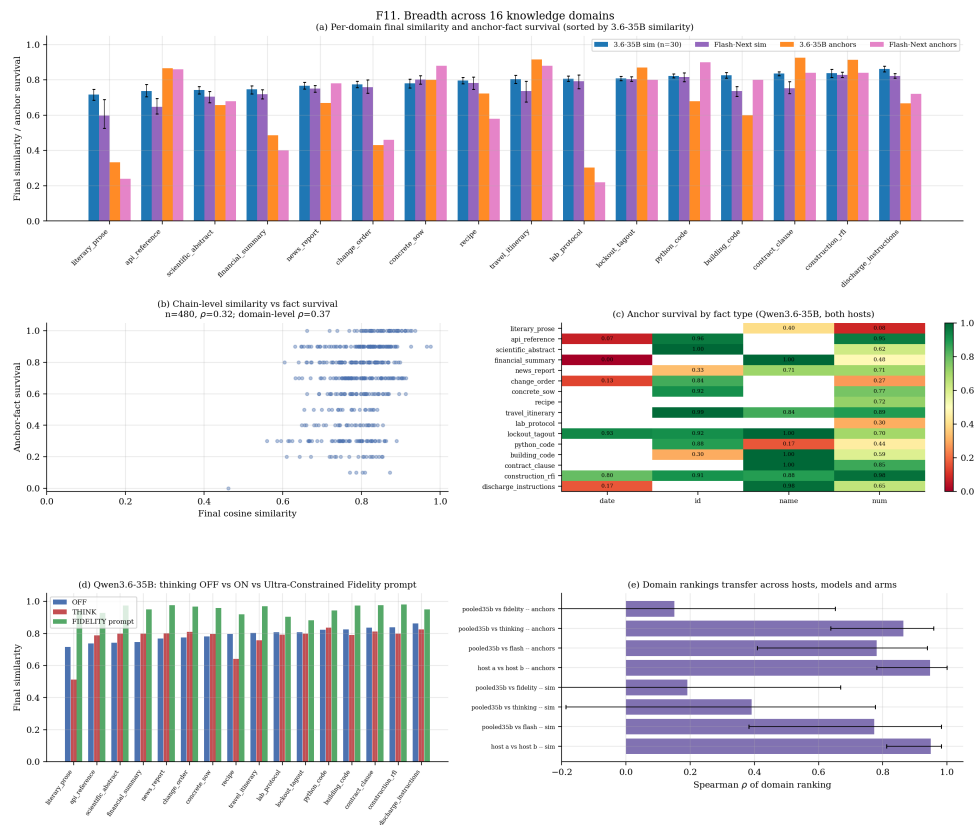


Figure 11: Sixteen domains on Qwen3.6-35B (30 samples per domain, both hosts) and Flash-Next (5). (a) Final similarity and anchor-fact survival per domain. (b) Similarity against survival per chain. (c) Survival by fact type. (d) Thinking off, thinking on, and the fidelity prompt per domain. (e) Domain rankings transfer across hosts, models and arms.

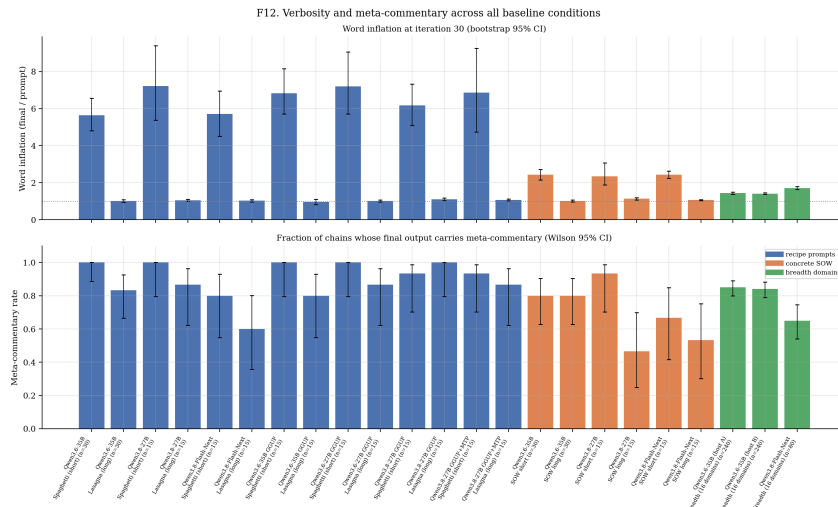


Figure 12: Word-count inflation and meta-commentary rate across every baseline condition: recipes, statements of work and the sixteen domains.

5 Discussion

What the similarity score measures. Three results converge. On the recipes, the short stimulus drifts far more than the long one while inflating five- to seven-fold and acquiring framing in nearly every chain. On the statements of work, similarity is anti-correlated with the survival of the numbers. Across sixteen domains, similarity varies by 0.14 while fact survival varies by 0.63, and the two are only weakly related. Cosine similarity to the original, the metric of Paper A and of most of the iterated-generation literature, responds first to length and genre change and only secondarily to information loss. Belem et al. [2026] found one axis of change it cannot see; Figure 12 shows that it over-reads another. The quantitative claims of Paper A about which model drifts least survive this—rankings replicate across metrics at $\rho \approx 0.8$ where we can compare them—but the interpretation of a value of 0.75 as “25% of the meaning lost” does not. We recommend that iterated-generation studies report a fact-survival measure beside any embedding score, and we release every chain’s text so that they can.

The hierarchy, revised again. Paper A ordered the factors governing drift as model family, then size to a threshold, then a discrete failure mode, then quantization format, then temperature and KV precision, with prompts as a multiplier. On the newer generations the ordering within a deployed model is: thinking mode (0.03–0.30, sign by model) and system prompt (0.24–0.43 spread) at the top, format-and-stack next (0 to 0.16), and temperature, seed and speculative decoding at or below the 0.05 noise floor. The two largest levers are the two that a pipeline designer sets in software after the model is chosen, and both are cheap to test at fifteen samples on these models.

Reasoning models in loops. The thinking results are, we think, the most consequential for practice. A reasoning pass makes Qwen3.8 a better paraphraser and Qwen3.6 a worse one; it prevents the copying attractor that otherwise ends most chains, so a loop containing a reasoning model does not settle; it deletes a bidder’s exclusions clause in most short statements of work while raising the similarity score; and on the previous generation it consumed the entire generation budget in a seventh of all chains. None of these effects is visible in a single-pass benchmark, and none is

predictable from the model’s single-pass quality. A pipeline that iterates over a reasoning model should be tested as a loop, with thinking on and off, on its own documents, with a fact-level check.

Recommendations.

1. Report iterated-generation results as samples, never as seeded replicates; fifteen chains per condition is a practical minimum.
2. Test thinking mode in both states on the deployed model and stimulus; do not assume its sign.
3. Use a fidelity instruction. One sentence lifted every domain tested to 0.88–0.98 with 97% of facts intact.
4. Measure fact survival, and check dates and quantities specifically; they are the first things lost.
5. Do not spend effort on seeds, greedy decoding or speculative decoding as drift controls; they do not change the outcome.
6. Benchmark the exact build and stack; the format effect is real on one of two models and absent on the other.

Limitations.

- **One family, four models.** Every configuration is a Qwen instruct model. The cross-generation trend is a within-family observation, and the Qwen3.5 anchors on the short stimulus rest on two or three clean chains.
- **Format confounds.** The GGUF arm differs from the NVFP4 arm in quantization, server, KV precision and GPU generation; the contrast bounds their combined effect.
- **Anchor facts are hand-chosen.** Ten per domain, chosen by the author to be checkable; survival is a lower bound on preservation (a paraphrased number counts as lost) and the domain prompts are synthetic documents rather than sampled ones.
- **Meta-commentary and genre drift are regex measures,** tuned on the outputs of these models; they will need re-tuning for others.
- **Small n in places.** Thinking-on breadth and the fidelity prompt use five samples per domain; the 100-iteration thinking runs on the Qwen3.8 models use five chains; the Qwen3.5 anchors are as stated above.
- **The sampler result is one model and one condition.** The presence-penalty effect was measured on Qwen3.6-35B at the single prompt with the highest token-cap rate; we expect it to generalise because the mechanism is repetition inside the reasoning block, but have not shown it on Qwen3.5 or on the other stacks.
- **Reasoning-token counts are estimated** for all but one deployment, from completion tokens minus the returned text, because only one server reported them.
- **Single embedding model.** We did not repeat the second- embedder validation of Paper A; the fact-survival metrics are the validation this paper offers instead.

6 Conclusion

Fast local inference let us replace assumptions with measurements. Seeds do not reproduce chains, so drift is a distribution; we measured it at fifteen to thirty chains per condition on eight configurations of four Qwen models. Thinking mode is the largest and least predictable factor, and reasoning models do not settle in a loop. Speculative decoding, greedy decoding and seeds change nothing. System-prompt rankings are stable and transferable within a generation, and one fidelity sentence makes a thirty-step chain nearly lossless in sixteen domains. Most chains end by copying their input verbatim. And the cosine-similarity score that this line of work has relied on responds to length and genre far more than to the facts a document carries, which are what a downstream reader needs: dates first, quantities next, names and identifiers last. The chain texts are released with this paper so that better measures can be applied to them.

References

- Alberto Acerbi and Joseph M. Stubbersfield. Large language models show human-like content biases in transmission chain experiments. *Proceedings of the National Academy of Sciences*, 120(44): e2313790120, 2023. doi: 10.1073/pnas.2313790120.
- Catarina G. Belem, Shang Wu, Hongyu Yao, Mark Steyvers, Sameer Singh, and Padhraic Smyth. From ‘may’ to ‘is’: Certainty distortion in language model rewriting. *arXiv preprint arXiv:2606.07951*, 2026.
- Levin Brinkmann, Fabian Baumann, Jean-François Bonnefon, Maxime Derex, Thomas F. Müller, Anne-Marie Nussberger, Agnieszka Czaplicka, Alberto Acerbi, Thomas L. Griffiths, Joseph Henrich, Joel Z. Leibo, Richard McElreath, Pierre-Yves Oudeyer, Jonathan Stray, and Iyad Rahwan. Machine culture. *Nature Human Behaviour*, 7(11):1855–1868, 2023. doi: 10.1038/s41562-023-01742-2.
- Charlie Chen, Sebastian Borgeaud, Geoffrey Irving, Jean-Baptiste Lespiau, Laurent Sifre, and John Jumper. Accelerating large language model decoding with speculative sampling. *arXiv preprint arXiv:2302.01318*, 2023.
- Sotirios Panagiotis Chytas and Vikas Singh. Concept attractors in LLMs and their applications. *arXiv preprint arXiv:2601.11575*, 2026.
- DeepSeek-AI. DeepSeek-V3 technical report. *arXiv preprint arXiv:2412.19437*, 2024.
- Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman and Hall/CRC, New York, 1993. doi: 10.1007/978-1-4899-4541-9.
- Georgi Gerganov and llama.cpp contributors. llama.cpp: LLM inference in C/C++. <https://github.com/ggml-org/llama.cpp>, 2023. Software repository; source of the GGUF format and the K-quant schemes (Q4_K_M, Q8_0).
- Fabian Gloeckle, Badr Youbi Idrissi, Baptiste Rozière, David Lopez-Paz, and Gabriel Synnaeve. Better & faster large language models via multi-token prediction. *arXiv preprint arXiv:2404.19737*, 2024.
- Henry Han, Xiyang Liu, Xiaodong Wang, Fei Han, and Xiaodong Li. The quantization trap: Breaking linear scaling laws in multi-hop reasoning. *arXiv preprint arXiv:2602.13595*, 2026.

- Horace He and Thinking Machines Lab. Defeating nondeterminism in LLM inference. <https://thinkingmachines.ai/blog/defeating-nondeterminism-in-llm-inference/>, sep 2025. Connectionism blog, Thinking Machines Lab.
- Coleman Hooper, Sehoon Kim, Hiva Mohammadzadeh, Michael W. Mahoney, Yakun Sophia Shao, Kurt Keutzer, and Amir Gholami. KVQuant: Towards 10 million context length LLM inference with KV cache quantization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2401.18079.
- Daniel Jaroslawicz, Brendan Whiting, Parth Shah, and Karime Maamari. How many instructions can LLMs follow at once? *arXiv preprint arXiv:2507.11538*, 2025.
- Simon Kirby, Hannah Cornish, and Kenny Smith. Cumulative cultural evolution in the laboratory: An experimental approach to the origins of structure in human language. *Proceedings of the National Academy of Sciences*, 105(31):10681–10686, 2008. doi: 10.1073/pnas.0707835105.
- Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP)*, pages 611–626, 2023. doi: 10.1145/3600006.3613165.
- Yaniv Leviathan, Matan Kalman, and Yossi Matias. Fast inference from transformers via speculative decoding. In *International Conference on Machine Learning (ICML)*, 2023. arXiv:2211.17192.
- Zirui Liu, Jiayi Yuan, Hongye Jin, Shaochen Zhong, Zhaozhuo Xu, Vladimir Braverman, Beidi Chen, and Xia Hu. KIVI: A tuning-free asymmetric 2bit quantization for KV cache. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2402.02750.
- Amr Mohamed, Mingmeng Geng, Michalis Vazirgiannis, and Guokan Shang. LLM as a broken telephone: Iterative generation distorts information. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025. arXiv:2502.20258.
- Sabbir Mollah, Rohit Gupta, Sirnam Swetha, Qingyang Liu, Ahnaf Munir, and Mubarak Shah. The telephone game: Evaluating semantic drift in unified models. *arXiv preprint arXiv:2509.04438*, 2025.
- NVIDIA. Qwen3.6-35b-a3b-nvfp4 model card. <https://huggingface.co/nvidia/Qwen3.6-35B-A3B-NVFP4>, 2026. Hugging Face model card, created 2026-05-27, Apache-2.0 license.
- NVIDIA, Felix Abecassis, Anjolie Agrusa, Dong Ahn, Jonah Alben, et al. Pretraining large language models with NVFP4. *arXiv preprint arXiv:2509.25149*, 2025.
- Jérémy Perez, Grgur Kovač, Corentin Léger, Cédric Colas, Gaia Molinaro, Maxime Derex, Pierre-Yves Oudeyer, and Clément Moulin-Frier. When LLMs play the telephone game: Cultural attractors as conceptual tools to evaluate LLMs in multi-turn settings. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2407.04503.
- Qwen Team. Qwen3.6-27b model card. <https://huggingface.co/Qwen/Qwen3.6-27B>, 2026a. Hugging Face model card, created 2026-04-21, Apache-2.0 license.
- Qwen Team. Qwen3.6-35b-a3b model card. <https://huggingface.co/Qwen/Qwen3.6-35B-A3B>, 2026b. Hugging Face model card, created 2026-04-15, Apache-2.0 license.

- Qwen Team. Qwen3.8-27b model card. <https://huggingface.co/Qwen/Qwen3.8-27B>, 2026c. Hugging Face model card, created 2026-08-05, Apache-2.0 license.
- Qwen Team. Qwen3.8-flash-next model card. <https://huggingface.co/Qwen/Qwen3.8-Flash-Next>, 2026d. Hugging Face model card, created 2026-08-24.
- Abhishek Rath. Agent drift: Quantifying behavioral degradation in multi-agent LLM systems over extended interactions. *arXiv preprint arXiv:2601.04170*, 2026.
- Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Yarin Gal, Nicolas Papernot, and Ross Anderson. The curse of recursion: Training on generated data makes models forget. *arXiv preprint arXiv:2305.17493*, 2023.
- Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson, and Yarin Gal. AI models collapse when trained on recursively generated data. *Nature*, 631(8022):755–759, 2024. doi: 10.1038/s41586-024-07566-y.
- James H. Smith. Semantic drift in iterated LLM paraphrase chains: Model architecture, scale, quantization, serving infrastructure, and prompt engineering. *arXiv preprint*, 2026. Companion paper; referred to as Paper A.
- Unsloth. Qwen3.6-35b-a3b-gguf model card. <https://huggingface.co/unsloth/Qwen3.6-35B-A3B-GGUF>, 2026a. Hugging Face model card, created 2026-04-16, Apache-2.0 license.
- Unsloth. Qwen3.8-27b-gguf model card. <https://huggingface.co/unsloth/Qwen3.8-27B-GGUF>, 2026b. Hugging Face model card, created 2026-08-13, Apache-2.0 license.
- Zhilin Wang, Yafu Li, Jianhao Yan, Yu Cheng, and Yue Zhang. Unveiling attractor cycles in large language models: A dynamical systems view of successive paraphrasing. *arXiv preprint arXiv:2502.15208*, 2025.
- Yanzhao Zhang, Mingxin Li, Dingkun Long, Xin Zhang, Huan Lin, Baosong Yang, Pengjun Xie, An Yang, Dayiheng Liu, Junyang Lin, Fei Huang, and Jingren Zhou. Qwen3 embedding: Advancing text embedding and reranking through foundation models. *arXiv preprint arXiv:2506.05176*, 2025a.
- Ziyang Zhang, Xinheng Ding, Jiayi Yuan, Rixin Liu, Huizi Mao, Jiarong Xing, and Zirui Liu. Deterministic inference across tensor parallel sizes that eliminates training-inference mismatch. *arXiv preprint arXiv:2511.17826*, 2025b.
- Shiju Zhao, Jiacheng Yang, Qihang Chen, Junhao Hu, Jiaqi Zheng, Guihai Chen, and Xusheng Chen. CoRun: Padding is simple and efficient for deterministic LLM inference. *arXiv preprint arXiv:2608.14376*, 2026.
- Lianmin Zheng, Liangsheng Yin, Zhiqiang Xie, Chuyue Sun, Jeff Huang, Cody Hao Yu, Shiyi Cao, Christos Kozyrakis, Ion Stoica, Joseph E. Gonzalez, Clark Barrett, and Ying Sheng. SGLang: Efficient execution of structured language model programs. *arXiv preprint arXiv:2312.07104*, 2024.

Table 7: System-prompt sensitivity. Spread is the best minus the worst prompt mean for that configuration and recipe; “chains” is the number of independent chains behind each prompt mean. The lower block gives the Spearman rank correlation of prompt orderings between every pair of configuration×recipe cells, with bootstrap 95% CIs.

Configuration	Recipe	prompts	chains	Spread	Best prompt	Worst prompt
3.6-35B (A)	spaghetti	20	10	0.306	100% Fidelity (0.971)	Visual Tutorial Style (0.665)
3.6-35B (A)	lasagna	20	5	0.323	100% Fidelity (0.968)	Multimodal Visual Focus (0.645)
3.6-35B (B)	spaghetti	20	5	0.298	Ultra-Constrained Fidelity (0.967)	Visual Tutorial Style (0.669)
3.6-35B (B)	lasagna	20	5	0.284	Precise Transmitter (0.966)	Visual Tutorial Style (0.682)
3.8-27B	spaghetti	20	5	0.266	Zero Fluff (0.937)	Baseline - None (0.672)
3.8-27B	lasagna	20	5	0.238	Balanced Clarity (0.965)	Multimodal Visual Focus (0.727)
Flash-Next	spaghetti	20	5	0.379	Ultra-Constrained Fidelity (0.950)	Visual Tutorial Style (0.572)
Flash-Next	lasagna	20	5	0.429	100% Fidelity (0.945)	Visual Tutorial Style (0.516)

Pair	n prompts	ρ	95% CI	p
3.6-35B (A) spag vs 3.6-35B (A) lasa	20	0.58	[0.12, 0.86]	0.0077
3.6-35B (A) spag vs 3.6-35B (B) spag	20	0.98	[0.91, 1.00]	<0.0001
3.6-35B (A) spag vs 3.6-35B (B) lasa	20	0.56	[0.15, 0.87]	0.0103
3.6-35B (A) spag vs 3.8-27B spag	20	0.83	[0.55, 0.94]	<0.0001
3.6-35B (A) spag vs 3.8-27B lasa	20	0.21	[-0.31, 0.68]	0.3695
3.6-35B (A) spag vs Flash-Next spag	20	0.90	[0.68, 0.97]	<0.0001
3.6-35B (A) spag vs Flash-Next lasa	20	0.15	[-0.42, 0.62]	0.5352
3.6-35B (A) lasa vs 3.6-35B (B) spag	20	0.56	[0.10, 0.84]	0.0106
3.6-35B (A) lasa vs 3.6-35B (B) lasa	20	0.95	[0.82, 0.98]	<0.0001
3.6-35B (A) lasa vs 3.8-27B spag	20	0.39	[-0.09, 0.74]	0.0923
3.6-35B (A) lasa vs 3.8-27B lasa	20	0.70	[0.37, 0.88]	0.0006
3.6-35B (A) lasa vs Flash-Next spag	20	0.49	[-0.04, 0.82]	0.0277
3.6-35B (A) lasa vs Flash-Next lasa	20	0.50	[-0.06, 0.85]	0.0261
3.6-35B (B) spag vs 3.6-35B (B) lasa	20	0.56	[0.12, 0.86]	0.0096
3.6-35B (B) spag vs 3.8-27B spag	20	0.83	[0.56, 0.93]	<0.0001
3.6-35B (B) spag vs 3.8-27B lasa	20	0.24	[-0.29, 0.67]	0.3038
3.6-35B (B) spag vs Flash-Next spag	20	0.91	[0.71, 0.98]	<0.0001
3.6-35B (B) spag vs Flash-Next lasa	20	0.17	[-0.40, 0.66]	0.4818
3.6-35B (B) lasa vs 3.8-27B spag	20	0.47	[-0.02, 0.80]	0.0369
3.6-35B (B) lasa vs 3.8-27B lasa	20	0.76	[0.47, 0.89]	0.0001
3.6-35B (B) lasa vs Flash-Next spag	20	0.54	[0.05, 0.88]	0.0140
3.6-35B (B) lasa vs Flash-Next lasa	20	0.54	[0.06, 0.88]	0.0137
3.8-27B spag vs 3.8-27B lasa	20	0.25	[-0.25, 0.67]	0.2855
3.8-27B spag vs Flash-Next spag	20	0.84	[0.61, 0.93]	<0.0001
3.8-27B spag vs Flash-Next lasa	20	0.21	[-0.32, 0.63]	0.3660
3.8-27B lasa vs Flash-Next spag	20	0.37	[-0.13, 0.77]	0.1053
3.8-27B lasa vs Flash-Next lasa	20	0.88	[0.67, 0.97]	<0.0001
Flash-Next spag vs Flash-Next lasa	20	0.33	[-0.23, 0.74]	0.1582

Table 8: Construction statement-of-work domain. Numeric survival is the fraction of the prompt’s distinct numeric tokens still present in the iteration-30 output; exclusions kept is the fraction of chains whose final output still contains the “by others” / “exclusion” language; genre drift is the fraction reframed as an email or memo. The lower block is the domain-neutral system-prompt subset on host B.

Configuration	Document	Think	n	Sim	Numeric	Inflation	Excl. kept	Genre drift
Qwen3.6-35B NVFP4	SOW short	OFF	30	0.751	1.00	2.43	0.87	0.63
Qwen3.6-35B NVFP4	SOW long	OFF	30	0.806	0.85	1.02	0.97	0.87
Qwen3.8-27B NVFP4	SOW short	OFF	15	0.743	1.00	2.34	0.80	0.60
Qwen3.8-27B NVFP4	SOW long	OFF	15	0.853	0.91	1.13	1.00	0.93
Qwen3.8-Flash-Next NVFP4	SOW short	OFF	15	0.733	1.00	2.42	0.67	1.00
Qwen3.8-Flash-Next NVFP4	SOW long	OFF	15	0.810	0.90	1.06	1.00	1.00
Qwen3.6-35B NVFP4	SOW short	ON	15	0.770	1.00	2.47	0.40	0.40
Qwen3.6-35B NVFP4	SOW long	ON	15	0.799	0.88	1.21	0.40	0.87
Qwen3.8-27B NVFP4	SOW short	ON	15	0.825	0.95	1.69	0.20	0.00
Qwen3.8-27B NVFP4	SOW long	ON	15	0.875	0.94	1.78	1.00	0.00
Qwen3.8-Flash-Next NVFP4	SOW short	ON	15	0.848	0.99	1.45	0.27	0.00
Qwen3.8-Flash-Next NVFP4	SOW long	ON	15	0.872	0.93	1.48	1.00	0.20

System prompt	Document	n	Sim	Numeric	Inflation	Excl. kept	Genre drift
Baseline - None	SOW short	5	0.750	1.00	2.94	1.00	0.60
Minimal Helpful	SOW short	5	0.760	1.00	2.79	0.80	1.00
Ultra-Constrained Fidelity	SOW short	5	0.964	1.00	1.06	1.00	0.00
Progressive Refinement	SOW short	5	0.893	0.98	1.04	0.40	0.00
Think-Step-by-Step	SOW short	5	0.846	0.91	1.22	0.00	0.00
Balanced Clarity	SOW short	5	0.840	1.00	1.56	1.00	0.00
Baseline - None	SOW long	5	0.776	0.80	0.92	1.00	1.00
Minimal Helpful	SOW long	5	0.821	0.89	1.17	1.00	0.80
Ultra-Constrained Fidelity	SOW long	5	0.982	1.00	1.05	1.00	0.00
Progressive Refinement	SOW long	5	0.885	0.91	0.80	1.00	0.00
Think-Step-by-Step	SOW long	5	0.919	0.91	0.93	1.00	0.00
Balanced Clarity	SOW long	5	0.883	0.92	1.10	1.00	0.00

Table 9: Breadth experiment, 16 knowledge domains, 30 iterations. The first block is Qwen3.6-35B NVFP4 with both hosts pooled and thinking OFF; THINK is thinking ON, FIDELITY is the domain-neutral Ultra-Constrained Fidelity system prompt, and Flash is Qwen3.8-Flash-Next NVFP4. Anchors = fraction of the 10 anchor facts surviving verbatim in the final output.

Domain	n	Sim	95% CI	Anchors	Infl.	Meta	Locked	THINK	FIDELITY	Flash
discharge_instructions	30	0.862	[0.846, 0.877]	0.67	1.14	0.70	0.83	0.826	0.949	0.823
construction_rfi	30	0.838	[0.814, 0.860]	0.91	1.34	0.93	0.83	0.798	0.980	0.829
contract_clause	30	0.836	[0.825, 0.847]	0.93	1.30	0.93	0.90	0.812	0.977	0.752
building_code	30	0.826	[0.810, 0.842]	0.60	1.22	0.87	0.83	0.790	0.973	0.737
python_code	30	0.823	[0.813, 0.833]	0.68	1.86	0.87	0.80	0.835	0.943	0.817
lockout_tagout	30	0.808	[0.795, 0.821]	0.87	1.27	0.77	0.83	0.798	0.882	0.805
lab_protocol	30	0.807	[0.792, 0.822]	0.30	1.62	0.70	0.57	0.792	0.905	0.792
travel_itinerary	30	0.804	[0.781, 0.827]	0.92	1.58	0.93	0.73	0.758	0.970	0.736
recipe	30	0.797	[0.779, 0.814]	0.72	1.53	0.80	0.70	0.642	0.920	0.783
concrete_sow	30	0.780	[0.756, 0.804]	0.80	1.65	0.83	0.83	0.796	0.958	0.802
change_order	30	0.774	[0.756, 0.792]	0.43	1.20	0.90	0.67	0.809	0.968	0.759
news_report	30	0.767	[0.749, 0.786]	0.67	1.27	0.80	0.93	0.802	0.976	0.751
financial_summary	30	0.746	[0.721, 0.767]	0.49	1.43	0.83	0.60	0.799	0.951	0.719
scientific_abstract	30	0.742	[0.721, 0.763]	0.66	1.39	0.83	0.83	0.799	0.975	0.705
api_reference	30	0.737	[0.704, 0.774]	0.87	1.61	0.87	0.80	0.788	0.928	0.647
literary_prose	30	0.716	[0.684, 0.748]	0.33	1.30	0.97	0.53	0.512	0.941	0.598

Table 10: Anchor-fact survival by fact type (Qwen3.6-35B NVFP4, both hosts pooled, thinking OFF, $n=30$ chains per domain). Each domain prompt carries exactly ten tagged anchor facts.

Domain	date	id	name	num
api_reference	0.07	0.96	—	0.95
building_code	—	0.30	1.00	0.59
change_order	0.13	0.84	—	0.27
concrete_sow	—	0.92	—	0.77
construction_rfi	0.80	0.91	0.88	0.98
contract_clause	—	—	1.00	0.85
discharge_instructions	0.17	—	0.98	0.65
financial_summary	0.00	—	1.00	0.48
lab_protocol	—	—	—	0.30
literary_prose	—	—	0.40	0.08
lockout_tagout	0.93	0.92	1.00	0.70
news_report	—	0.33	0.71	0.71
python_code	—	0.88	0.17	0.44
recipe	—	—	—	0.72
scientific_abstract	—	1.00	—	0.62
travel_itinerary	—	0.99	0.84	0.89
All domains	0.35	0.81	0.80	0.63

A Stimuli

A.1 Recipes

The two recipes are those of Paper A. **Spaghetti**: “Step 1: Boil water. Step 2: Add pasta for 8 minutes. Step 3: Drain and serve with sauce.” **Lasagna**: the 21-step beef lasagna recipe of Paper A, Appendix A, sent with steps 12–17 compressed into one line, exactly as in that paper.

A.2 Construction statements of work

concrete short.

Summary Statement of Work, Trade: Concrete, Section 03 30 00 Cast-in-Place Concrete. Scope: 1. Furnish and place 185 CY of 4,000 psi normal-weight concrete for spread footings and grade beams, 4 in slump, 6 percent air. 2. Furnish and install 9.2 tons of ASTM A615 Grade 60 reinforcing steel with 3 in cover at earth-formed surfaces. 3. Provide edge formwork, 7-day moist curing, and one set of four ASTM C31 test cylinders per 50 CY placed, broken per ASTM C39 at 7 and 28 days. Major exclusions: excavation, backfill, and dewatering by the Sitework contractor.

concrete long.

Statement of Work, Trade: Concrete (Cast-in-Place), Building B Warehouse Expansion. Spec sections used: 03 20 00 Concrete Reinforcing, 03 30 00 Cast-in-Place Concrete, 03 35 00 Concrete Finishing, 07 26 00 Vapor Retarders. The Concrete subcontractor shall furnish all labor, materials, equipment, and supervision for the following scope. Foundations: 1. Spread footings, 24 each, 6 ft x 6 ft x 18 in, 4,000 psi concrete, 48 CY total. 2. Continuous wall footings, 620 LF, 2 ft wide x 12 in deep, 4,000 psi, 46 CY. 3. Foundation walls, 620 LF, 8 in thick x 4 ft high, 4,000 psi, 61 CY, formed both faces. 4. Loading dock pits, 4 each, 10 ft x 8 ft x 4 ft deep with 8 in walls and slab, 5,000 psi, 22 CY, with embedded steel angles at all edges. 5. Anchor bolts: install 96 each 1 in diameter x 24 in ASTM F1554 Grade 36 anchor bolts per templates furnished by the Steel Erector. Slabs: 6. Slab-on-grade, 42,000 SF, 6 in thick, 4,500 psi, 778 CY, placed over 15-mil vapor retarder per ASTM E1745 Class A. 7. Equipment pads, 6 each, 4 ft x 6 ft x 6 in, 4,000 psi, 3 CY, with 3/4 in chamfered edges. 8. Sawcut control joints at 15 ft maximum spacing each way, 1/4 of slab depth, cut within 12 hours of finishing. 9. Slab finish: hard steel trowel, floor flatness FF 35 and floor levelness FL 25 per ASTM E1155, tested within 72 hours of placement. Reinforcing: 10. Slab reinforcing: #4 bars at 18 in on center each way, ASTM A615 Grade 60, on chairs at 3 ft on center. 11. Footing and wall reinforcing: 14.5 tons ASTM A615 Grade 60, 3 in cover against earth, 2 in cover at formed surfaces; furnish mill certificates. Materials and quality: 12. Mix designs: maximum water-cement ratio 0.45, 4 in slump plus or minus 1 in, air entrainment 6 percent plus or minus 1.5 percent for exterior concrete, Type II cement, 25 percent Class F fly ash replacement permitted. 13. Curing: dissipating-resin curing compound at 200 SF per gallon within 30 minutes of final finishing; moist-cure footings and walls for 7 days. 14. Cold weather: maintain concrete at 55 degrees F minimum for 3 days after placement when ambient temperature is below 40 degrees F, per ACI 306. 15. Tolerances per ACI 117: footings plus 2 in or minus 1/2 in in plan; walls plus or minus 1/4 in in 10 ft. 16. Testing: one set of four cylinders per 100 CY or fraction thereof per day per mix, ASTM C31 field cured, ASTM C39 breaks at 7 and 28 days, slump and air per ASTM C143 and ASTM C231 at each set. 17. Submittals within 14 calendar days of award: concrete mix designs, reinforcing shop drawings, vapor retarder and curing compound product data, mill certificates. Schedule: 18. Footings complete within 15 working days of mobilization; slab-on-grade placed in 6 pours of approximately 7,000 SF each; 28-day design strength required before steel erection loads. Exclusions (by others): 19. Excavation, backfill, dewatering, and compacted subgrade by the Sitework contractor. 20. The 4 in granular capillary break under slabs by the Sitework contractor. 21. Testing agency fees

paid by the Owner; anchor bolt templates furnished by the Steel Erector; dumpster for concrete debris provided by the General Contractor. Alternates: Alternate 1: substitute 6 x 6 W2.9 x W2.9 welded wire reinforcement for the #4 slab bars, deduct \$18,400. Base bid: \$1,284,600 including a 5 percent contingency.

A.3 Sixteen-domain breadth prompts

Each prompt is followed by its ten anchor facts (type: text).

recipe.

Chicken tikka masala for 4 people. Cut 1.5 lb boneless chicken thighs into 1-inch pieces and marinate 30 minutes in 1 cup plain yogurt, 2 tablespoons lemon juice, 2 teaspoons garam masala, 1 teaspoon turmeric, and 1 teaspoon salt. Heat 2 tablespoons ghee in a large skillet over medium-high heat and sear the chicken 4 minutes per side until charred at the edges, then set aside. In the same pan cook 1 diced onion for 5 minutes, add 4 minced garlic cloves and 1 tablespoon grated ginger, and cook 1 minute more. Stir in 2 tablespoons tomato paste and 1 teaspoon smoked paprika, then pour in one 14-oz can of crushed tomatoes and simmer 10 minutes. Add 3/4 cup heavy cream and the chicken, and simmer 8 minutes until the sauce thickens. Finish with 2 tablespoons chopped cilantro. Serve over basmati rice with warm naan.

Anchors: num: "1.5 lb"; num: "30 minutes"; num: "2 teaspoons garam masala"; num: "1 teaspoon turmeric"; num: "4 minutes per side"; num: "4 minced garlic cloves"; num: "14-oz"; num: "10 minutes"; num: "3/4 cup heavy cream"; num: "8 minutes"

lab protocol.

Protocol: plasmid miniprep by alkaline lysis. Pellet 3 mL of overnight E. coli culture at 8,000 x g for 2 minutes and discard the supernatant. Resuspend the pellet in 250 uL of Buffer P1 containing RNase A at 100 ug/mL and vortex until no clumps remain. Add 250 uL of Buffer P2 (200 mM NaOH, 1 percent SDS), invert the tube 6 times, and incubate no longer than 5 minutes at room temperature. Neutralise with 350 uL of Buffer N3, invert 6 times, and centrifuge at 13,000 rpm for 10 minutes. Transfer the cleared lysate to a silica spin column, spin 60 seconds, and discard the flow-through. Wash with 500 uL of Buffer PB and then 750 uL of Buffer PE, spinning 60 seconds after each wash and once more dry for 1 minute. Elute in 50 uL of Buffer EB warmed to 50 degrees C, letting it stand 1 minute before the final spin. Expected yield is 5 to 15 ug; verify by measuring A260/A280, which should be between 1.8 and 2.0.

Anchors: num: "3 mL"; num: "8,000 x g"; num: "250 uL"; num: "100 ug/mL"; num: "200 mM NaOH"; num: "350 uL"; num: "13,000 rpm"; num: "750 uL"; num: "50 degrees C"; num: "1.8 and 2.0"

lockout tagout.

Lockout/tagout procedure LOTO-114 for Hydraulic Press 7 in Building C. Only authorised employees trained under OSHA 29 CFR 1910.147 may perform this procedure. Step 1: Notify the shift supervisor at extension 4471 and all affected employees that the press will be shut down. Step 2: Stop the press using the operator stop button and confirm the ram is fully retracted. Step 3: Open the main electrical disconnect MCC-3, breaker 12, and apply a red personal lock and tag. Step 4: Close the hydraulic supply valve HV-7A and apply a second lock. Step 5: Bleed residual pressure at the gauge port until the gauge reads 0 psi; the accumulator holds up to 3,000 psi and must be fully vented. Step 6: Attempt a restart to verify zero energy state, then return the controls to off. Step 7: Perform the work. Step 8: Remove tools, replace guards, confirm all personnel are clear, and remove locks in reverse order. Each lock stays with its owner; a lock may be cut only by the plant manager with a written release. Review this procedure annually, next due 15 March 2027.

Anchors: id: "LOTO-114"; name: "Hydraulic Press 7"; id: "29 CFR 1910.147"; num: "extension 4471"; id: "MCC-3"; num: "breaker 12"; id: "HV-7A"; num: "3,000 psi"; name: "plant manager"; date: "15 March 2027"

concrete sow.

Statement of Work, Trade: Concrete, Section 03 30 00 Cast-in-Place Concrete, Building B Warehouse Expansion. Scope: 1. Spread footings, 24 each, 6 ft x 6 ft x 18 in, 4,000 psi concrete, 48 CY total. 2. Foundation walls, 620 LF, 8 in thick x 4 ft high, 4,000 psi, 61 CY, formed both faces. 3. Slab-on-grade, 42,000 SF, 6 in thick, 4,500 psi, 778 CY, over a 15-mil vapor retarder per ASTM E1745. 4. Reinforcing: 14.5 tons of ASTM A615 Grade 60 steel with 3 in cover against earth; slab bars #4 at 18 in on center each way. 5. Mix design: maximum water-cement ratio 0.45, 4 in slump, 6 percent air for exterior concrete. 6. Sawcut control joints at 15 ft maximum spacing within 12 hours of finishing. 7. Testing: one set of four ASTM C31 cylinders per 100 CY, broken at 7 and 28 days. 8. Submittals within 14 calendar days of award. Exclusions: excavation, backfill, and dewatering by the Sitework contractor; anchor bolt templates by the Steel Erector. Base bid: \$1,284,600 including a 5 percent contingency.

Anchors: id: "03 30 00"; num: "24 each"; num: "4,000 psi"; num: "620 LF"; num: "42,000 SF"; num: "778 CY"; id: "ASTM E1745"; num: "14.5 tons"; num: "0.45"; num: "\$1,284,600"

construction rfi.

Request for Information RFI-047, Project: Maple Street Elementary Addition, Date: 12 August 2026, From: Harlan Mechanical, To: Kessler Architects, Response required by 19 August 2026. Subject: Rooftop unit RTU-3 curb height conflict. Drawing M-401 shows RTU-3 on a 14-inch roof curb at grid line D-7, but structural drawing S-201 detail 5 shows a 24-inch curb to clear the tapered insulation, which is 10 inches thick at that location per A-501. Specification Section 23 74 13 paragraph 2.4.B requires the curb to be factory-fabricated and shipped with the unit, and the unit was released to the manufacturer on 30 July 2026 with the 14-inch curb. Question 1: Which curb height governs? Question 2: If 24 inches is required, will the Owner accept a field-fabricated 12-inch extension welded to the factory curb, or must the unit be re-released at a cost of \$6,850 and 6 weeks of delay? Question 3: Does the condensate drain routing on M-401 change if the curb height changes? Contractor recommendation: accept the field extension with flashing per SMACNA Figure 8-2. Please respond in writing; work at grid D-7 is on hold.

Anchors: id: "RFI-047"; name: "Harlan Mechanical"; name: "Kessler Architects"; date: "19 August 2026"; id: "RTU-3"; id: "M-401"; id: "D-7"; num: "14-inch"; num: "24-inch"; num: "\$6,850"

change order.

Change Order Request COR-019, Project: Riverside Clinic Renovation, Contract No. 2026-114, Date: 3 September 2026. Reason: unforeseen asbestos-containing floor mastic discovered under the existing vinyl tile in Rooms 112 through 118, confirmed by laboratory report EnvTest-88213 dated 28 August 2026. Scope: abatement of 2,340 square feet of mastic by a licensed abatement subcontractor, air monitoring, and disposal at a permitted landfill. Cost breakdown: abatement labor, 4 workers for 6 days at \$68 per hour, \$13,056; abatement materials and encapsulant, \$2,410; air monitoring and clearance testing, \$3,200; disposal, 3 lined containers at \$850 each, \$2,550; subcontractor overhead and profit at 15 percent, \$3,182; general contractor markup at 10 percent, \$2,440. Total this change order: \$26,838. Schedule impact: 8 working days added to the Substantial Completion date, moving it from 14 November 2026 to 26 November 2026. This proposal is valid for 30 days. Owner approval is required before abatement begins; the flooring installer has been placed on standby.

Anchors: id: "COR-019"; id: "2026-114"; num: "2,340 square feet"; id: "EnvTest-88213"; num: "\$68 per hour"; num: "\$13,056"; num: "\$3,200"; num: "15 percent"; num: "\$26,838"; date: "26 November 2026"

contract clause.

Article 11, Indemnification. To the fullest extent permitted by law, the Subcontractor shall indemnify, defend, and hold harmless the Contractor, the Owner, and their respective officers, agents, and employees from and against all claims, damages, losses, and expenses, including reasonable attorneys' fees, arising out of or resulting from the performance of the Subcontract Work, but only to the extent caused by the negligent acts or omissions of the Subcontractor, its sub-subcontractors, or anyone for whose acts they may be liable. This obligation shall not be construed to negate or reduce any other right of indemnity that would otherwise exist. The Subcontractor shall maintain commercial general liability insurance of not less than \$2,000,000 per occurrence and \$4,000,000 aggregate, naming the Contractor and Owner as additional insureds on a primary and non-contributory basis, with a waiver of subrogation. Certificates of insurance shall be delivered within 10 days of execution and not less than 30 days before any policy expires. Notice of any claim shall be given in writing within 14 days of the Subcontractor first becoming aware of it. This Article survives termination of the Subcontract for a period of 3 years.

Anchors: name: "Article 11"; name: "Subcontractor"; name: "Owner"; num: "\$2,000,000"; num: "\$4,000,000"; name: "additional insureds"; name: "waiver of subrogation"; num: "10 days"; num: "14 days"; num: "3 years"

building code.

Section 1005, Means of Egress Sizing (excerpt). 1005.3.1 Stairways: the capacity of stairways shall be calculated at 0.3 inch of width per occupant, except that in buildings equipped throughout with an automatic sprinkler system and an emergency voice alarm system, 0.2 inch per occupant is permitted. 1005.3.2 Other egress components: 0.2 inch per occupant, reduced to 0.15 inch where the sprinkler and voice alarm exception applies. 1005.7 Encroachment: doors opening into the path of egress travel shall not reduce the required width by more than 7 inches when fully open, and by not more than one-half at any point during the swing. Minimum stair width is 44 inches where the occupant load served exceeds 50, and 36 inches otherwise. Handrails may project not more than 4.5 inches into the required width on each side. Corridors serving an occupant load of 50 or more shall be not less than 44 inches wide; in Group E occupancies with more than 100 occupants, 72 inches. Where two exits are required, each shall be sized for not less than 50 percent of the total occupant load.

Anchors: id: "1005.3.1"; num: "0.3 inch"; num: "0.2 inch"; num: "0.15 inch"; num: "7 inches"; num: "44 inches"; num: "36 inches"; num: "4.5 inches"; name: "Group E"; num: "72 inches"

discharge instructions.

Discharge instructions for Robert Alvarez, age 67, discharged 4 September 2026 after treatment for community-acquired pneumonia. Medications: amoxicillin-clavulanate 875 mg by mouth every 12 hours for 7 more days, take with food; azithromycin 250 mg once daily for 4 more days; continue lisinopril 10 mg every morning; acetaminophen 650 mg every 6 hours as needed for fever, not to exceed 3,000 mg in 24 hours. Do not take ibuprofen while on lisinopril without asking your doctor. Use the incentive spirometer 10 breaths every hour while awake. Drink at least 2 liters of fluid daily unless told otherwise. Call the clinic at 555-0142 or go to the emergency department if your temperature exceeds 101.5 degrees F, if you become short of breath at rest, or if you cough up blood. Follow-up appointment with Dr. Priya Natarajan on 11 September 2026 at 9:30 AM; bring this sheet and all medication bottles. A repeat chest X-ray is scheduled for 6 weeks after discharge to confirm the infiltrate has cleared. No driving for 48 hours after the last dose of the cough medicine containing codeine.

Anchors: name: "Robert Alvarez"; num: "875 mg"; num: "every 12 hours"; num: "250 mg"; num: "10 mg"; num: "3,000 mg"; num: "555-0142"; num: "101.5 degrees F"; name: "Dr. Priya Natarajan"; date: "11 September 2026"

python code.

Python module for rate limiting. `def token_bucket(capacity: int = 100, refill_rate: float = 10.0):` creates a closure holding two variables, `tokens` initialised to `capacity` and `last_refill` initialised to `time.monotonic()`. The inner function `allow(cost: int = 1) -> bool` first computes `elapsed = time.monotonic() - last_refill`, then sets `tokens = min(capacity, tokens + elapsed * refill_rate)` and updates `last_refill`. If `tokens >= cost` it subtracts `cost` and returns `True`; otherwise it returns `False` without changing `tokens`. The function raises `ValueError` if `capacity <= 0` or `refill_rate <= 0`. A second helper, `retry_after(cost: int) -> float`, returns `max(0.0, (cost - tokens) / refill_rate)`, the seconds until enough tokens will exist. Both helpers are returned as a tuple `(allow, retry_after)`. Usage: `allow, retry_after = token_bucket(capacity=50, refill_rate=5.0)`; if not `allow(3)`: `sleep(retry_after(3))`. The module depends only on the standard library `time` module and is thread-unsafe by design; callers must wrap it in a `threading.Lock` when sharing across threads. Version 1.4.2, licensed under the MIT License.

Anchors: id: "token_bucket"; num: "capacity: int = 100"; num: "refill_rate: float = 10.0"; id: "time.monotonic()"; id: "allow(cost: int = 1)"; id: "ValueError"; id: "retry_after"; id: "threading.Lock"; num: "1.4.2"; name: "MIT License"

api reference.

Endpoint: `POST /v2/invoices`. Creates an invoice for a customer. Authentication: Bearer token with the `invoices:write` scope. Request body (JSON): `customer_id` (string, required, format `cus_` followed by 14 characters); `currency` (string, required, ISO 4217 code such as `USD`); `line_items` (array, required, 1 to 250 objects each with `description`, `quantity` as an integer, and `unit_amount` in minor units); `due_date` (string, optional, ISO 8601 date, defaults to 30 days after creation); `metadata` (object, optional, up to 50 keys). Response 201 Created returns the invoice object with `id` (format `inv_` plus 14 characters), `status` set to `draft`, and `total` computed as the sum of `quantity` times `unit_amount`. Errors: 400 `invalid_request` when a required field is missing; 402 `payment_required` when the account is past due; 404 `not_found` when `customer_id` does not exist; 409 `conflict` when an `Idempotency-Key` header was already used with a different body; 429 `rate_limited` when more than 100 requests per minute are sent. Idempotency keys are retained for 24 hours. This endpoint was introduced in API version 2025-11-01 and replaces `POST /v1/invoices`, which is deprecated and will be removed on 1 June 2027.

Anchors: id: "POST /v2/invoices"; id: "invoices:write"; id: "customer_id"; id: "ISO 4217"; num: "250"; num: "201 Created"; num: "402"; num: "409"; num: "24 hours"; date: "1 June 2027"

scientific abstract.

Background: Whether a 12-week structured walking programme reduces blood pressure in adults with stage 1 hypertension remains uncertain. Methods: We randomised 248 participants aged 45 to 70 years at 6 primary care clinics to either supervised walking for 40 minutes 5 days per week or usual care. The primary outcome was change in systolic blood pressure at 12 weeks. Results: 231 participants (93 percent) completed follow-up. Mean systolic pressure fell by 7.8 mmHg in the walking group versus 2.1 mmHg with usual care, a between-group difference of 5.7 mmHg (95 percent confidence interval 3.4 to 8.0; $p < 0.001$). Diastolic pressure fell by 3.2 mmHg more in the walking group ($p = 0.004$). Adherence averaged 4.3 sessions per week. Two participants in the walking group reported musculoskeletal injuries; no serious adverse events occurred. The effect was larger in participants with a baseline body mass index above 30 (interaction $p = 0.03$). Conclusions: A supervised walking programme produced a clinically meaningful reduction in systolic blood pressure over 12 weeks. Trial registration NCT05873310.

Anchors: num: "12-week"; num: "248"; num: "45 to 70"; num: "40 minutes"; num: "93 percent"; num: "7.8 mmHg"; num: "5.7 mmHg"; num: " $p < 0.001$ "; num: "4.3 sessions"; id: "NCT05873310"

financial summary.

Northwind Logistics Inc. reported results for the second quarter ended 30 June 2026. Revenue was \$412.6 million, up 9.4 percent from \$377.1 million in the prior-year quarter, driven by a 14 percent increase in cross-border freight volume. Gross margin expanded to 23.8 percent from 21.9 percent as fuel costs declined. Operating income was \$38.2 million, and net income was \$27.5 million, or \$1.12 per diluted share, compared with \$0.89 a year earlier. Adjusted EBITDA reached \$61.7 million. Operating cash flow for the six months was \$88.3 million; capital expenditures were \$31.0 million, mostly for 140 new tractors. The company ended the quarter with \$96.4 million in cash and \$210 million of undrawn revolver capacity, and repurchased 1.1 million shares for \$52 million. Management raised full-year revenue guidance to a range of \$1.62 billion to \$1.66 billion and reaffirmed an expected effective tax rate of 24 percent. The board declared a quarterly dividend of \$0.18 per share payable on 15 October 2026 to holders of record on 30 September 2026.

Anchors: name: "Northwind Logistics"; num: "\$412.6 million"; num: "9.4 percent"; num: "23.8 percent"; num: "\$27.5 million"; num: "\$1.12"; num: "\$61.7 million"; num: "1.1 million shares"; num: "\$0.18"; date: "15 October 2026"

news report.

PORTLAND, Oregon, 5 September 2026. A 6.1-magnitude earthquake struck off the Oregon coast at 4:47 a.m. local time on Friday, the U.S. Geological Survey said, centred about 62 miles west of Newport at a depth of 10 kilometres. No tsunami warning was issued. Oregon Emergency Management reported minor damage to a bridge on U.S. Route 20 near Toledo and a power outage affecting roughly 8,400 customers in Lincoln County, most restored by 9 a.m. Two people were treated for minor injuries at Samaritan Pacific Communities Hospital, a spokesperson said. Governor Elena Whitcombe said state inspectors would examine 14 coastal bridges over the weekend. Amtrak suspended Cascades service between Portland and Eugene for 3 hours while tracks were inspected. Seismologist Daniel Okafor of Oregon State University said the quake occurred on the Cascadia subduction zone's outer fault system and that aftershocks above magnitude 4 were possible for several days. The last quake of comparable size in the region was a 6.3 event in 2004.

Anchors: num: "6.1"; num: "4:47 a.m."; name: "Newport"; num: "10 kilometres"; id: "U.S. Route 20"; num: "8,400"; name: "Elena Whitcombe"; num: "14 coastal bridges"; name: "Daniel Okafor"; num: "2004"

travel itinerary.

Itinerary for Maria Chen, confirmation code QX7R4M. Monday 21 September 2026: depart Boston Logan Terminal B on JetBlue flight B6 1523 at 7:15 AM, arrive Chicago O'Hare Terminal 5 at 9:02 AM, seat 14C. Pick up a mid-size rental car from Hertz, reservation H8823391, at the O'Hare rental center. Check in after 3:00 PM at the Kimpton Gray Hotel, 122 West Monroe Street, Chicago, reservation 44192077, 2 nights, king room, \$289 per night. Tuesday 22 September: client meeting at 10:00 AM with Ardent Partners, 233 South Wacker Drive, Suite 4100, contact Luis Ortega, 312-555-0177; dinner reservation at 7:30 PM for 4 at Girl & the Goat. Wednesday 23 September: return the car by 12:00 PM, depart O'Hare on B6 1530 at 2:40 PM, arrive Boston at 6:05 PM. Total airfare \$486.40 charged to the corporate card ending 7731. Travel insurance policy TI-90315 covers trip cancellation up to \$2,500.

Anchors: name: "Maria Chen"; id: "QX7R4M"; id: "B6 1523"; num: "7:15 AM"; num: "14C"; id: "H8823391"; name: "122 West Monroe Street"; id: "44192077"; name: "Luis Ortega"; num: "\$486.40"

literary prose.

The lighthouse keeper’s daughter kept a ledger of the fog. Not the weather service kind, with its dutiful columns of visibility and wind, but a private accounting of how the grey came in: some mornings like a tide, filling the harbour from the bottom up until the masts of the fishing boats stood in it like reeds; other mornings all at once, a door closing on the world. Her father said fog was only water that had forgotten which way was down. She wrote that in the ledger too. In October the fog stayed nine days without lifting, and she learned the sound of the town without its shapes: the bakery shutter at six, the school bell that seemed to ring from underwater, the horn on the point answering itself every thirty seconds like a man who has lost the thread of an argument. When it finally cleared she found the harbour exactly as it had been, and was surprised to feel disappointed, as though the fog had promised to take something with it when it went and had failed to keep its word.

Anchors: name: “lighthouse keeper’s daughter”; name: “ledger”; name: “harbour”; name: “fishing boats”; name: “forgotten which way was down”; name: “October”; num: “nine days”; name: “bakery shutter”; name: “school bell”; num: “thirty seconds”

B Additional figures

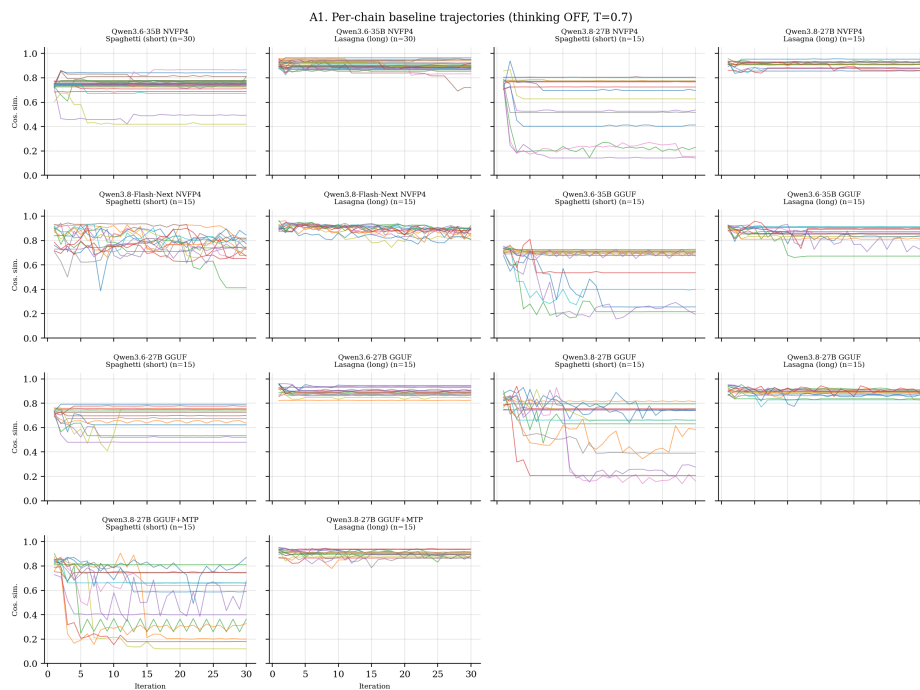


Figure 13: Per-chain baseline trajectories for every configuration.

C Reproducibility and compute

Hardware. Qwen3.6-35B NVFP4 was served by vLLM on one NVIDIA RTX 5090 (32 GB) on each of two machines (hosts A and B). Qwen3.8-27B NVFP4 was served by vLLM with tensor parallelism across two NVIDIA RTX 5070 Ti (16 GB each). Qwen3.8-Flash-Next NVFP4 was served by SGLang on one NVIDIA RTX PRO 6000 Blackwell (96 GB). All GGUF builds were served

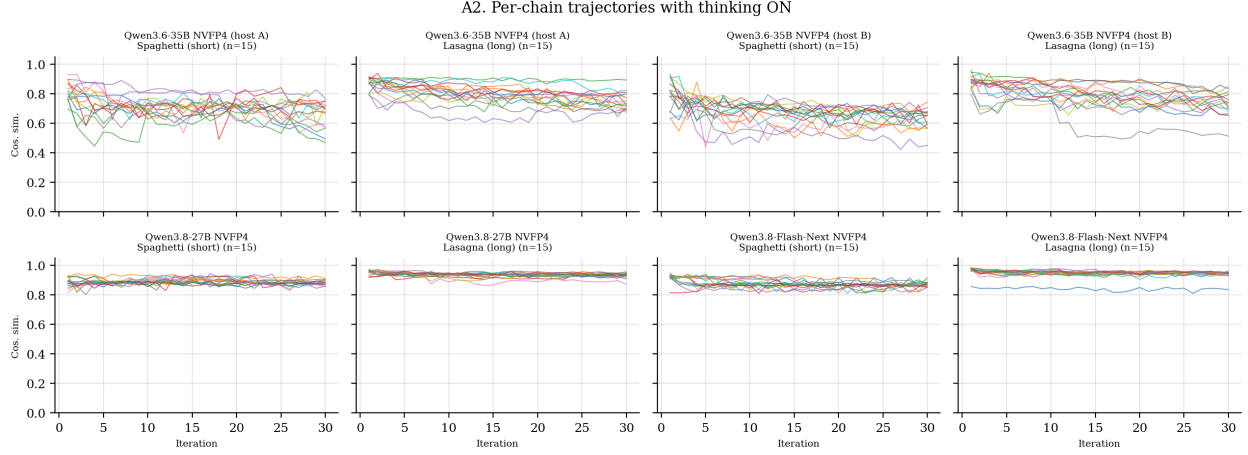


Figure 14: Per-chain trajectories with thinking on.

by llama.cpp with tensor splitting across two NVIDIA Quadro P5000 (16 GB each), one model resident at a time. Embeddings were computed by Qwen3-Embedding-0.6B served by vLLM on an RTX 5070 Ti. The GGUF runs of Paper A used Ollama on the RTX PRO 6000; the change of server and GPU generation between the two papers’ GGUF arms is part of the format-and-stack confound noted in Section 4.4.

Software. vLLM (NVFP4 builds; FlashInfer MoE backend, fp8 KV cache, prefix caching, reasoning parser `qwen3`, thinking disabled by default and enabled per request via `chat_template_kwargs`), SGLang (Flash-Next; ModelOpt FP4, reasoning parser `auto`), llama.cpp router (GGUF; q8_0 KV cache, reasoning off; `draft-mtp` with draft length 2 for the MTP arm). The generation budget was 16,384 tokens and the request timeout 600s throughout.

Scale. 2,717 chains, 87,110 model calls, run between 6 and 7 September 2026; the Blackwell-hosted conditions completed in under 24 hours of wall-clock time and the llama.cpp conditions in a further 16 hours.

Availability. The runner, the analysis script that produces every figure and table here, the prompt files with their anchors, and the full per-iteration chain texts are at <https://github.com/jhsmith409/telephone-semantic-drift>, with the chain data attached to the repository’s data release as a checksummed archive.